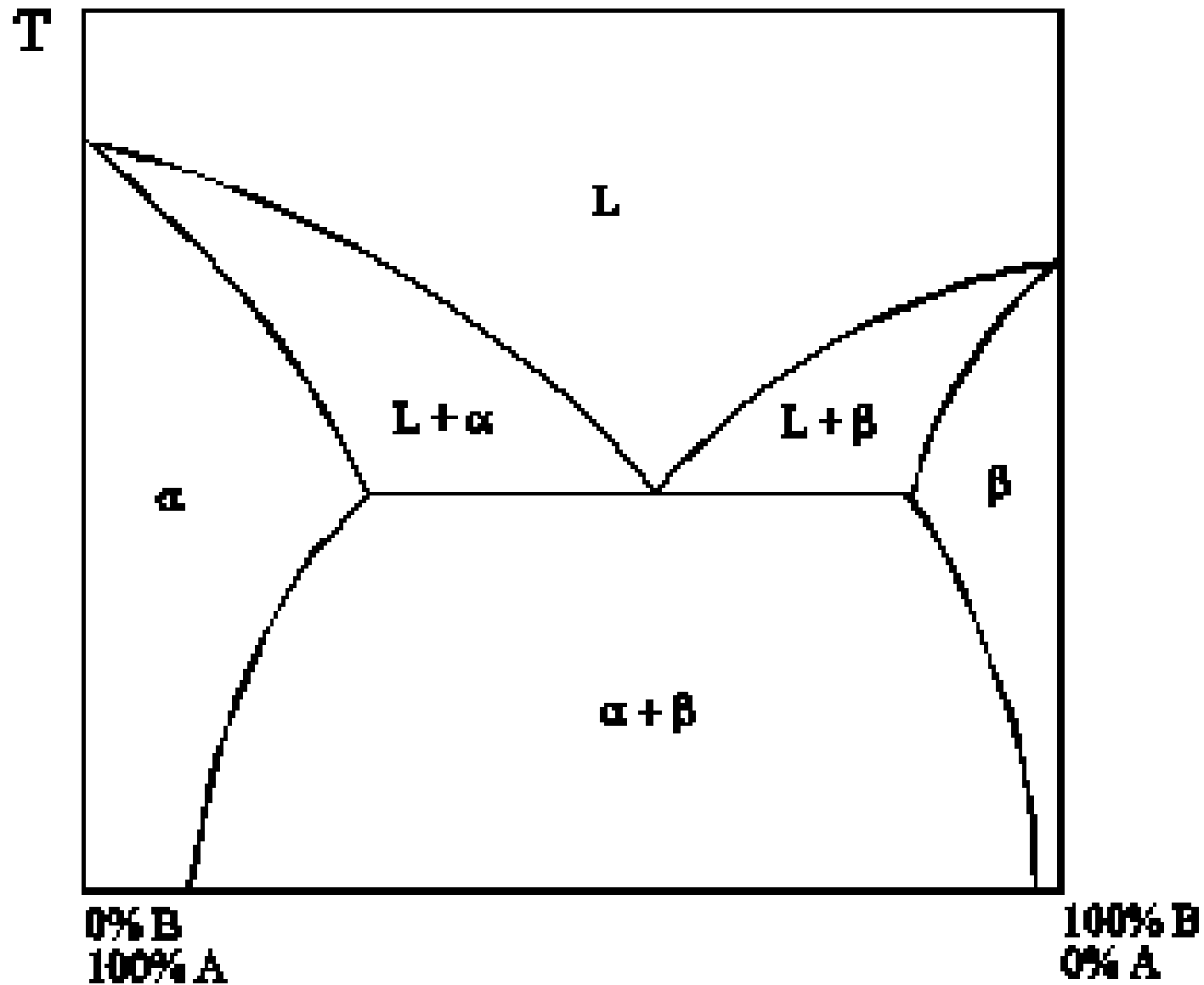




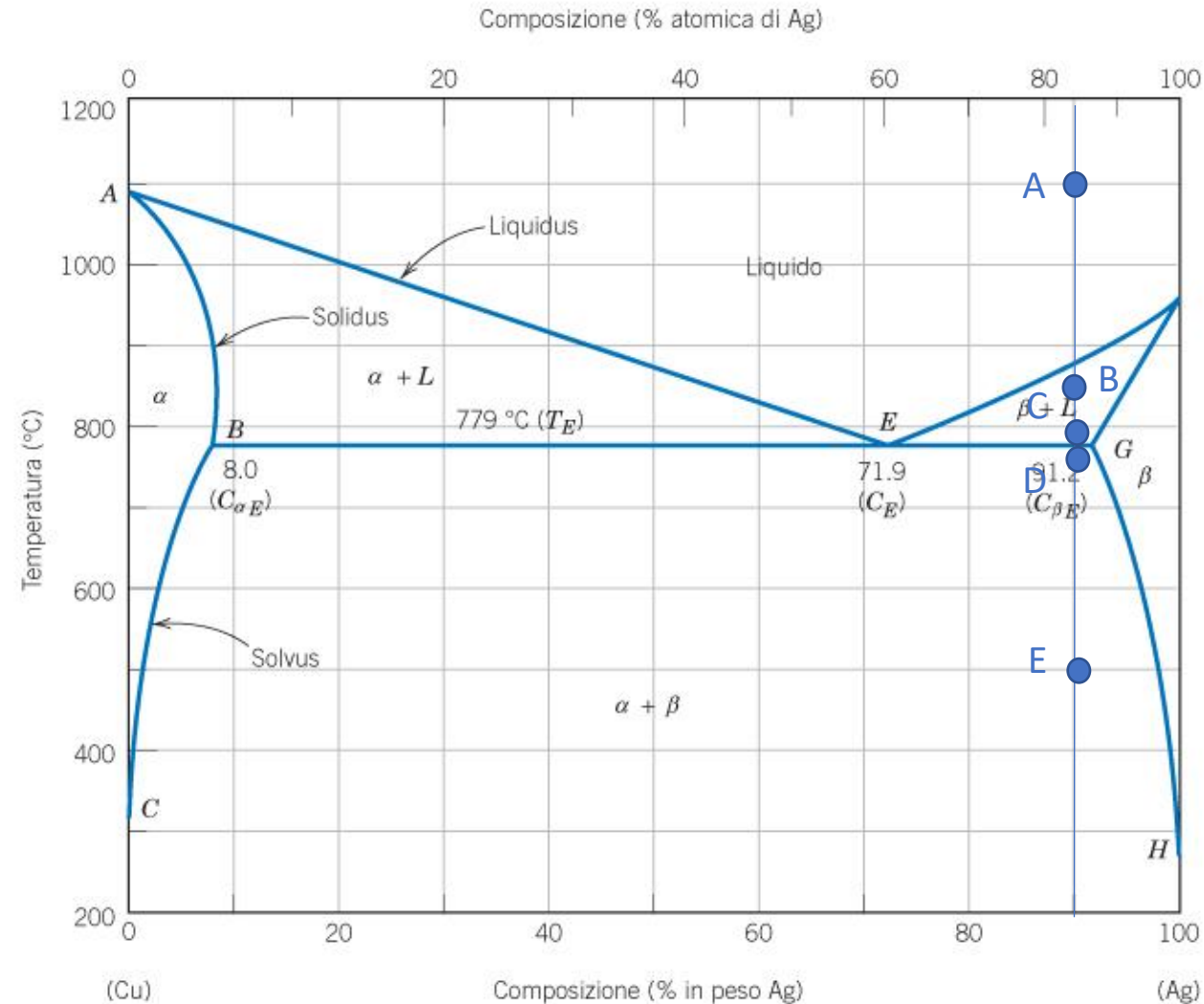
# STM Exercise 7

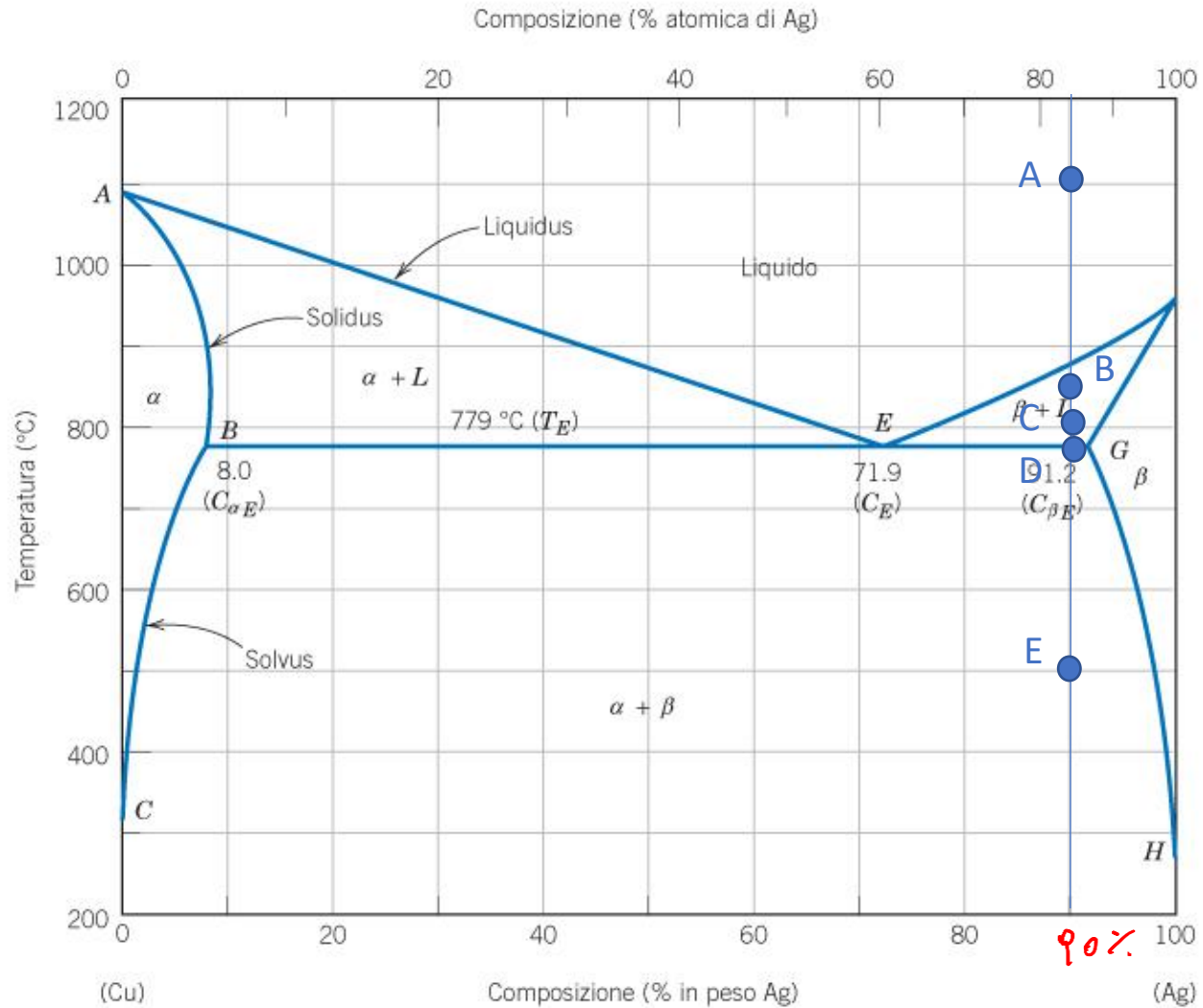
Phase Diagrams  
(partial solubility in the solid state)

Complete solubility in the liquid state,  
partial solubility in the solid state



**Exercise 1:** determine the phase present, their composition and relative amounts in the points **A**, **B**, **C** (at the beginning of the eutectic transformation), **D** (at the end of the eutectic transformation) and **E** indicated in the phase diagram. Schematize the microstructure in each point.



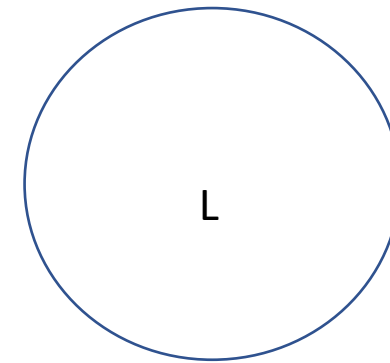


### Point A:

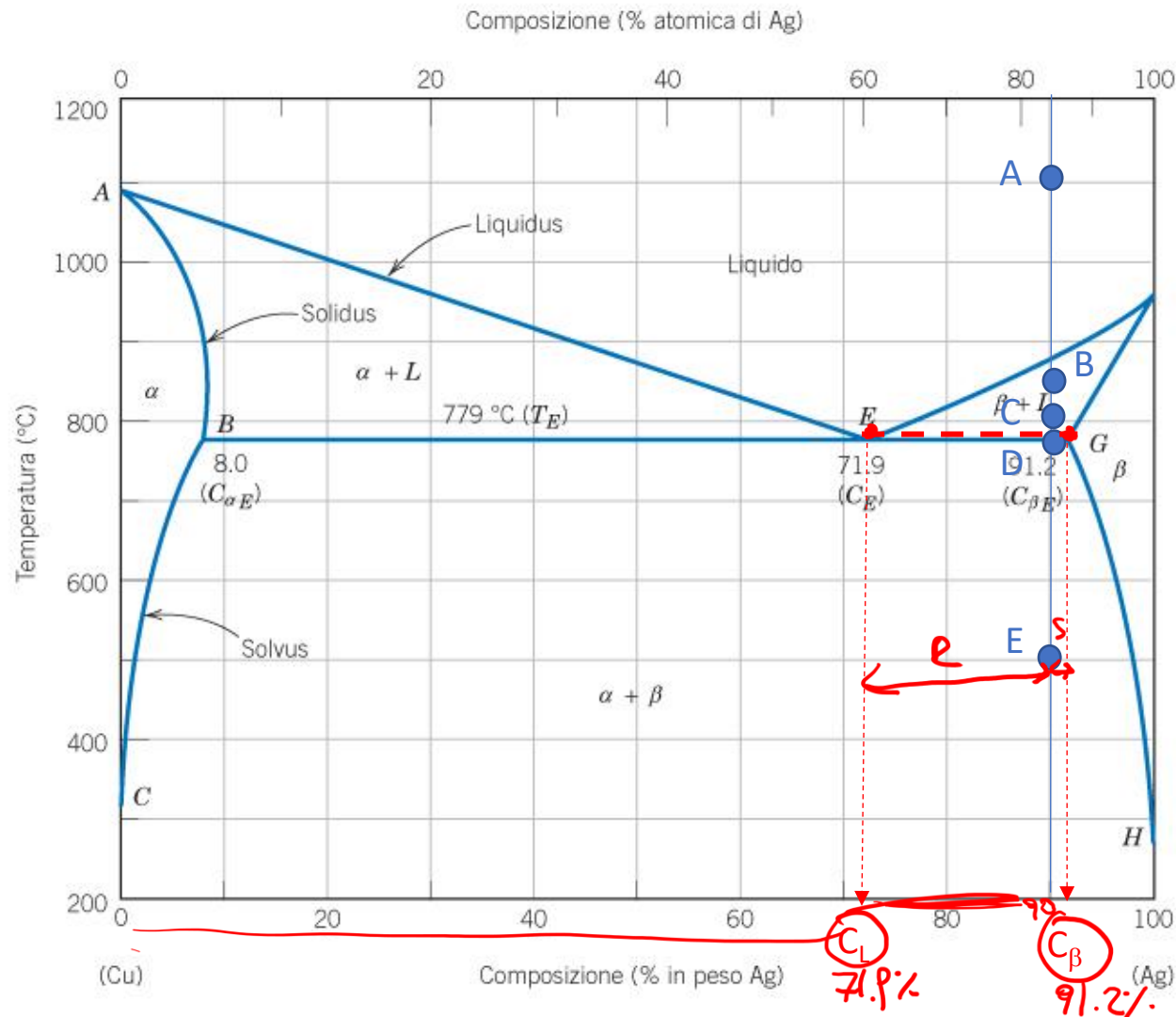
**Phases:** 1 phase liquid

**Composition:** 90%Ag, 10%Cu

**Relative amount:** 100% liquid phase







### Point C:

**Phases:** 2 phases solid ( $\beta$ ) + liquid (L)

**Composition:** 2 phases region → horizontal rule

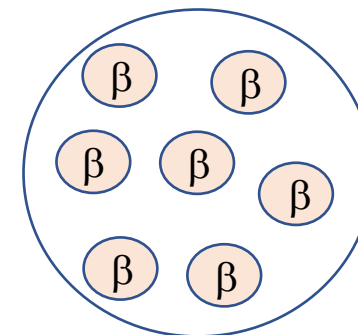
**Composition of liquid (L):**  $C_L = 71.9\% \text{ Ag}$ ,  $28.1\% \text{ Cu}$

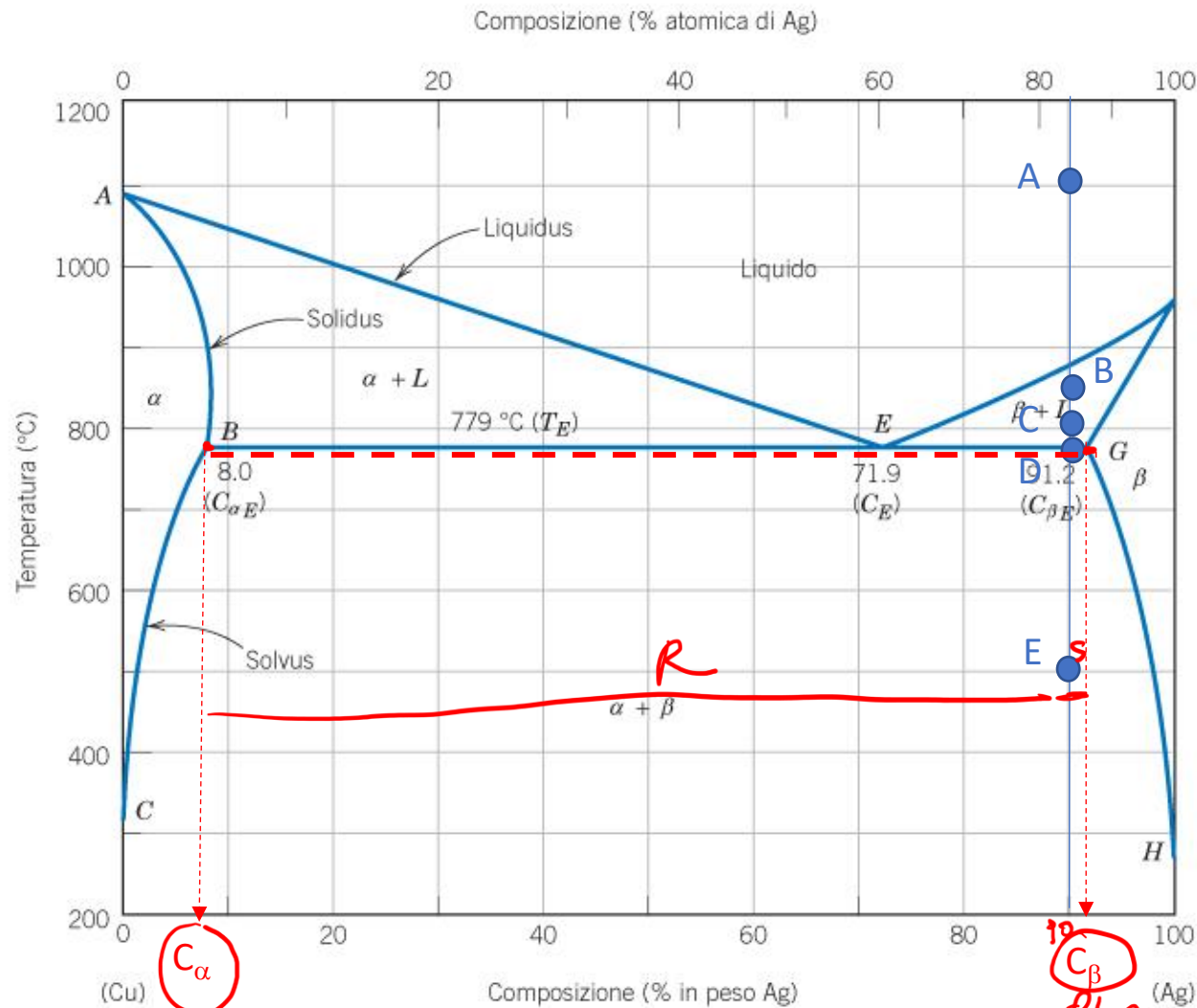
**Composition of solid ( $\beta$ ):**  $C_{\beta} = 91.2\% \text{ Ag}$ ,  $8.8\% \text{ Cu}$

**Relative amount:** 2 phases region → lever rule

$$\% \beta = \frac{R}{R+S} = \frac{90-71.9}{91.2-71.9} \times 100 = 93.8\%$$

$$\% L = \frac{S}{S+R} = \frac{91.2-90}{91.2-71.9} \times 100 = 6.2\%$$





### Point D:

**Phases:** 2 phases solid ( $\alpha$ ) + solid ( $\beta$ )

**Composition:** 2 phases region → horizontal rule

**Composition of solid ( $\alpha$ ):**  $C_{\alpha} = 8\% \text{Ag}, 92\% \text{Cu}$

**Composition of solid ( $\beta$ ):**  $C_{\beta} = 91.2\% \text{Ag}, 8.8\% \text{Cu}$

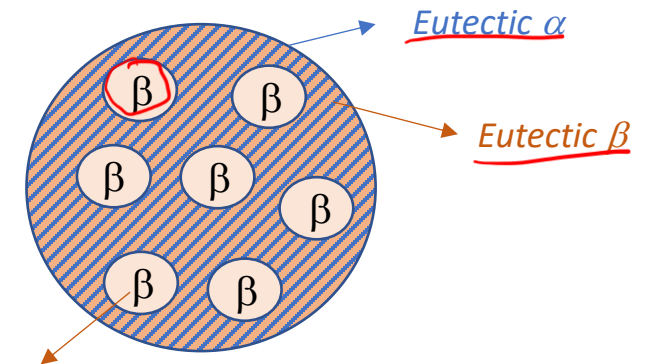
**Relative amount:** 2 phases region → lever rule

$$\% \alpha = \frac{(91.2 - 90)}{(91.2 - 8)} \times 100 = 1.4\%$$

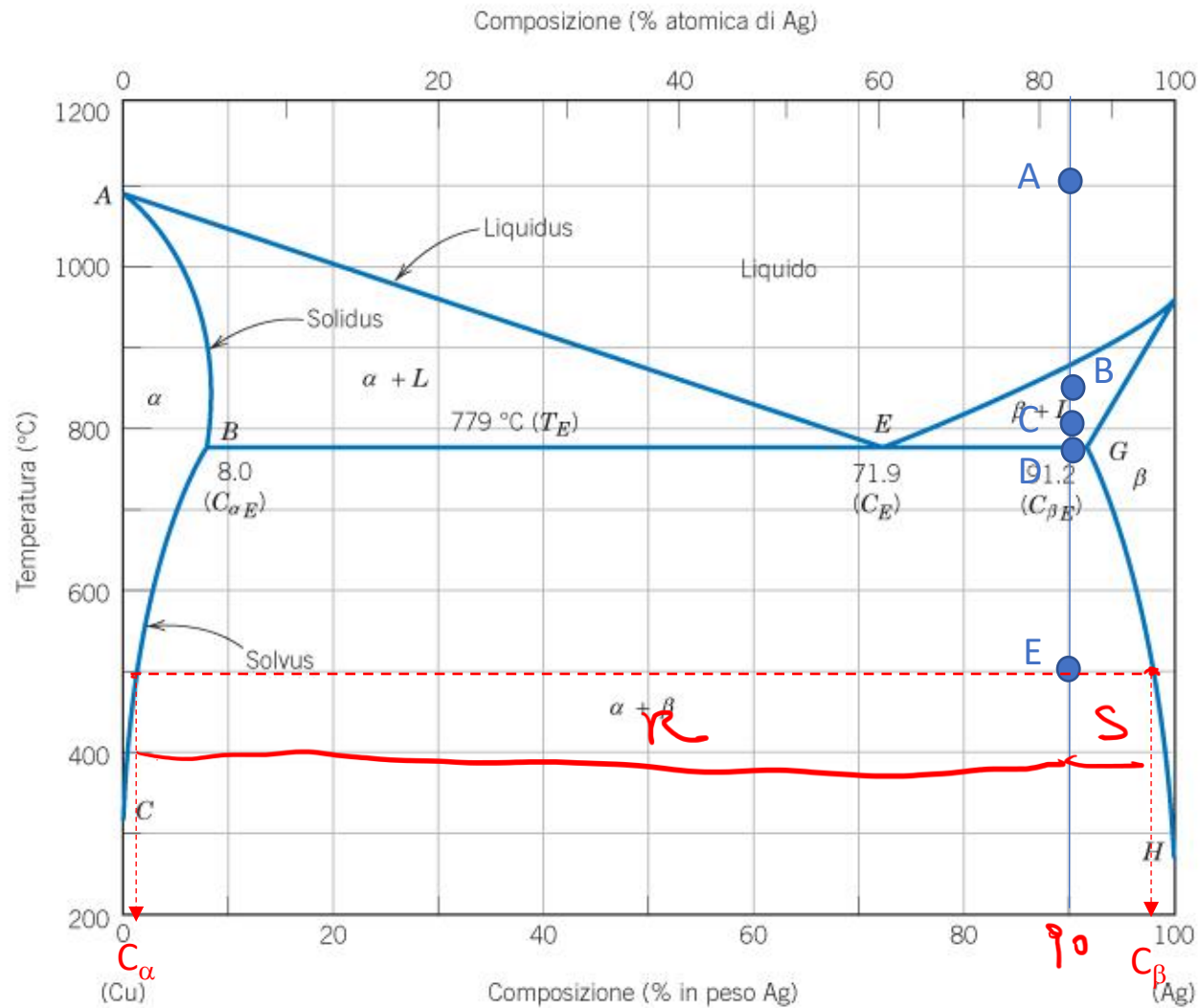
$$\% \beta = \frac{(90 - 8)}{(91.2 - 8)} \times 100 = 98.6\%$$

*S* *S+R*

*R* *S+R*



Primary/pro-eutectic  $\beta$



### Point E:

**Phases:** 2 phases solid (α) + solid (β)

**Composition:** 2 phases region → horizontal rule

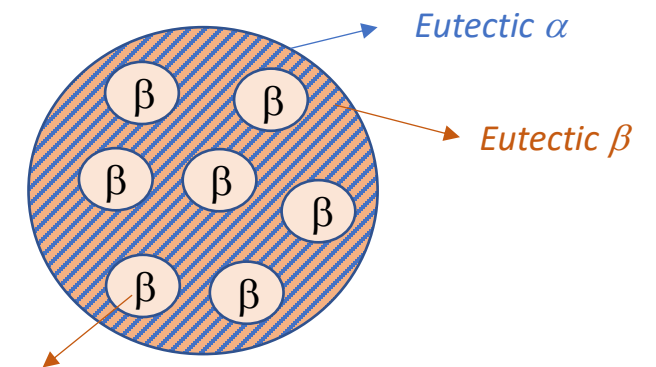
**Composition of solid (α):** C<sub>α</sub> = 2%Ag, 98%Cu

**Composition of solid (β):** C<sub>β</sub> = 98%Ag, 2%Cu

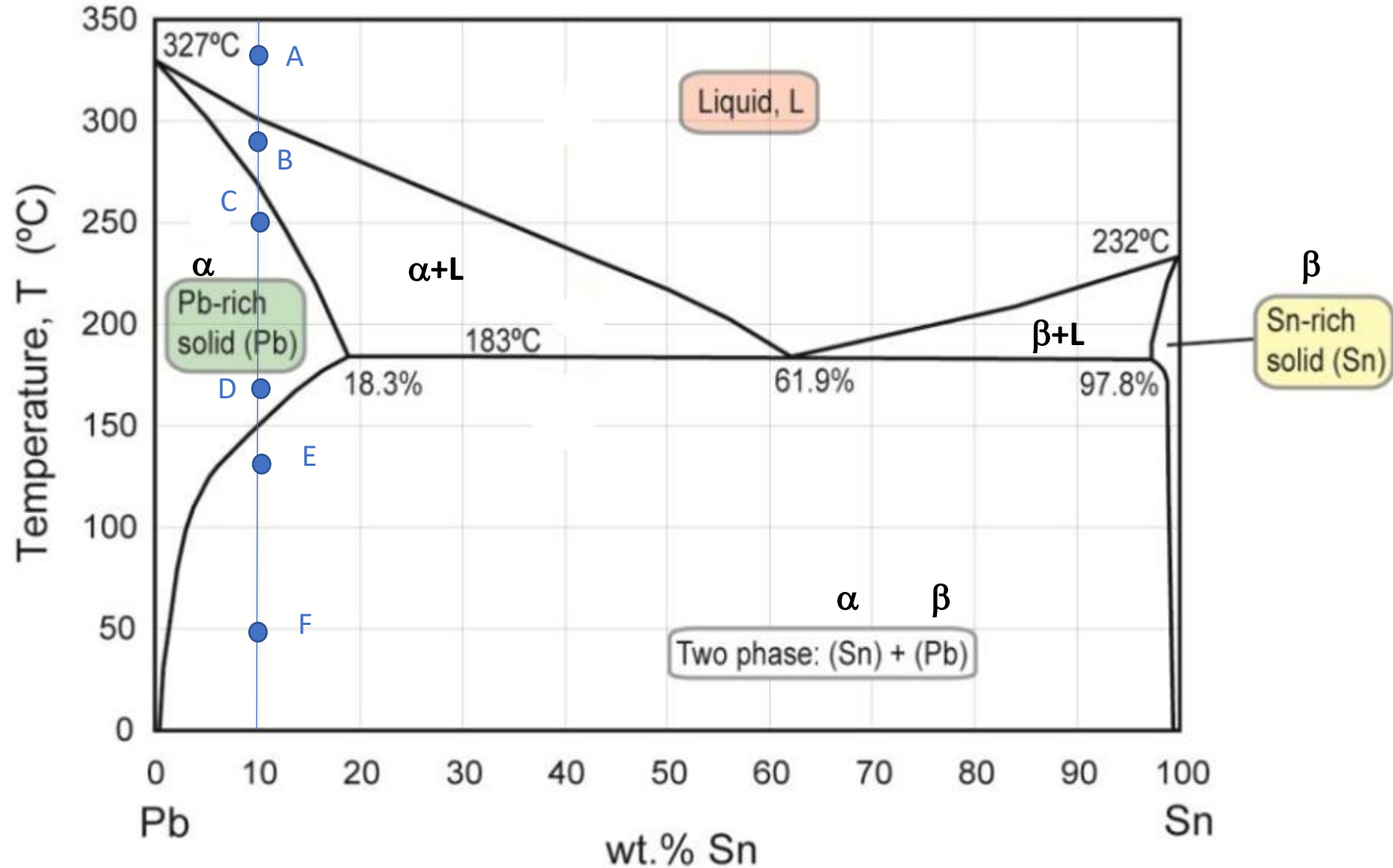
**Relative amount:** 2 phases region → lever rule

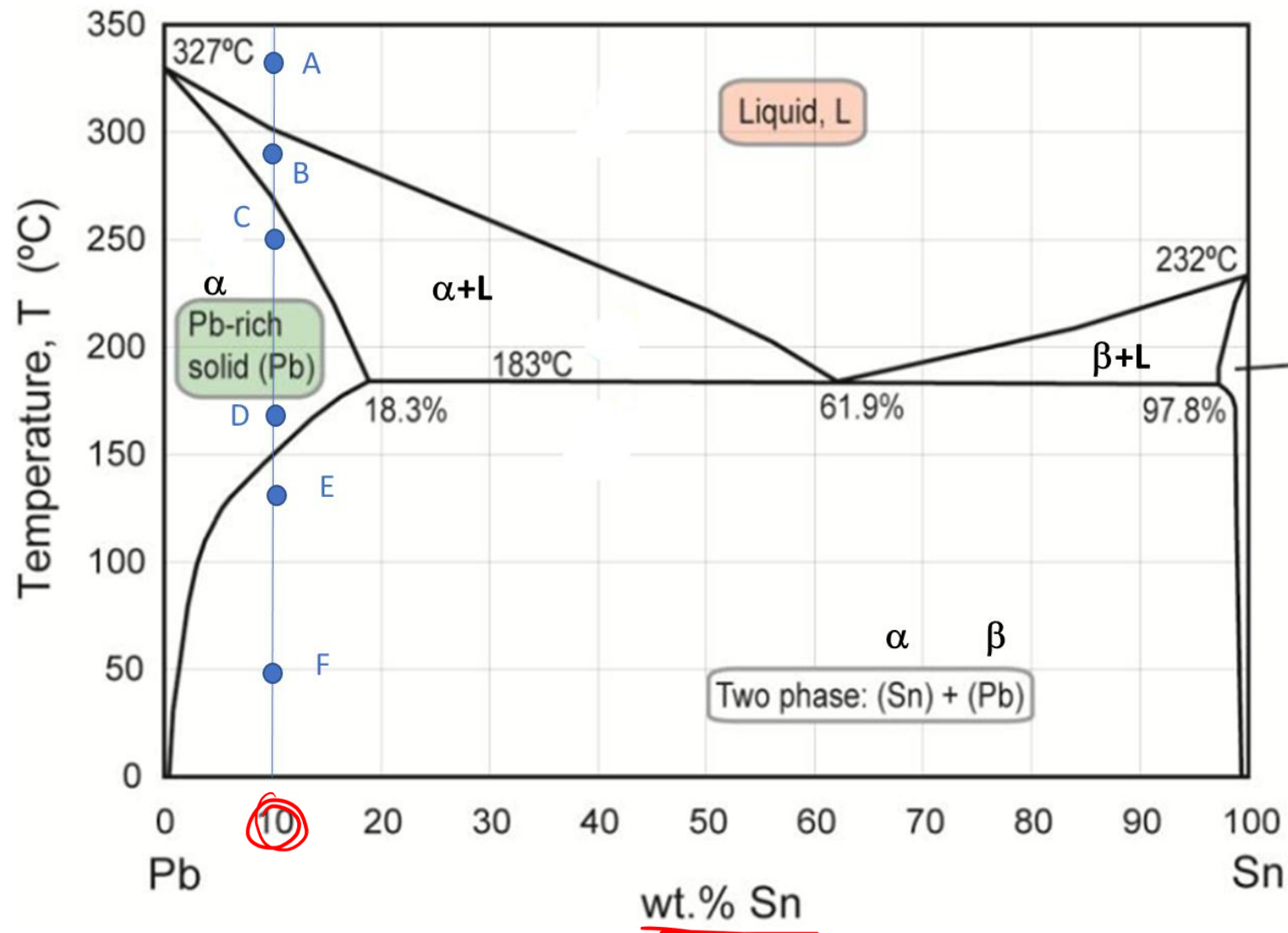
$$-\% \alpha = (98 - 90) / (98 - 2) * 100 = 8.3\%$$

$$\% \beta = (90 - 2) / (98 - 2) * 100 = 91.7\%$$



**Exercise 2:** determine the phase present, their composition and relative amounts in the points A, B, C, D, E and F indicated in the phase diagram. Schematize the microstructure in each point.



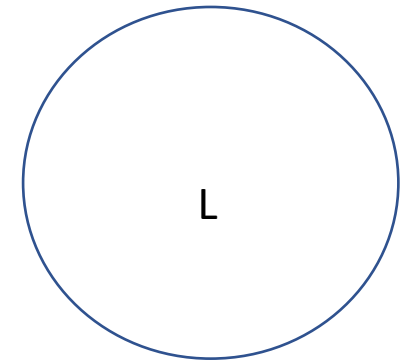


**Point A:**

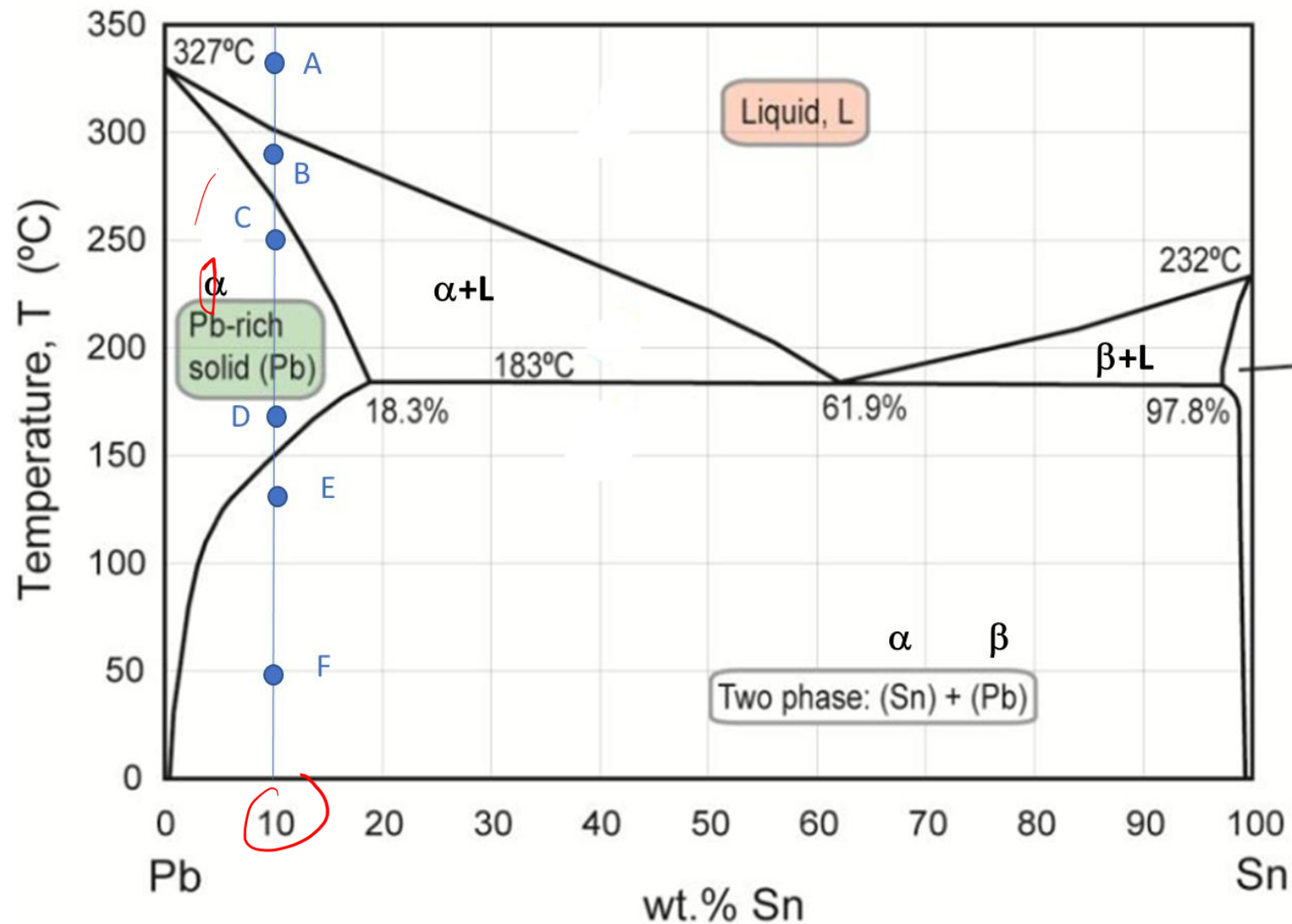
**Phases:** 1 phase liquid

**Composition:** 10% Sn, 90% Pb

**Relative amount:** 100% liquid phase





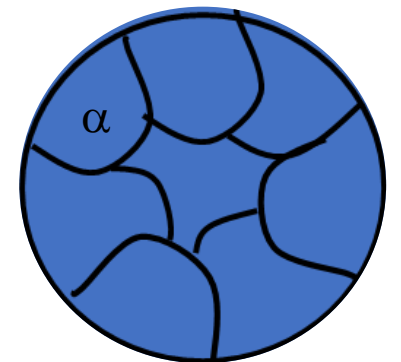


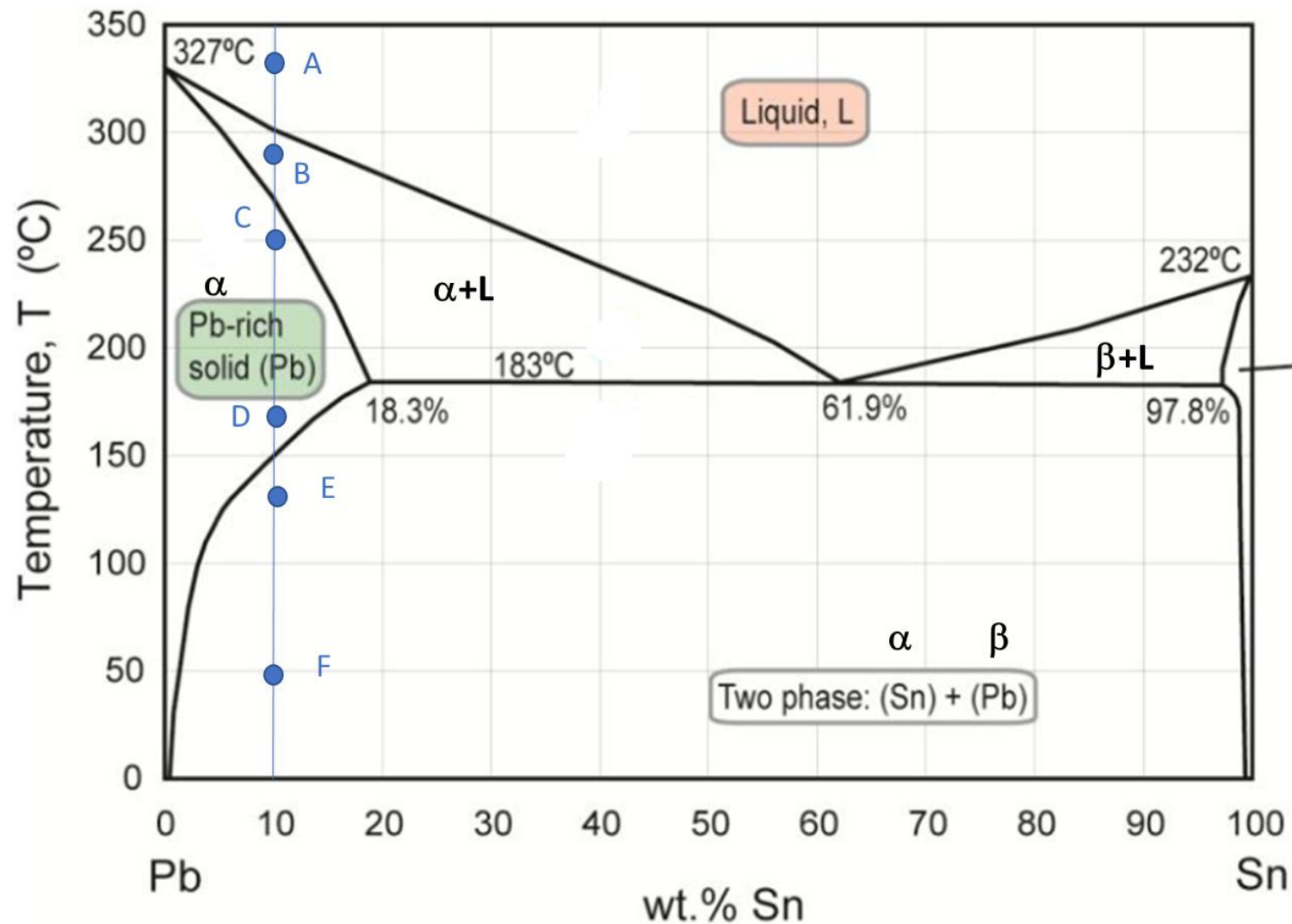
### Point C:

**Phases:** 1 phase solid ( $\alpha$ )

**Composition:** 10% Sn, 90% Pb

**Relative amount:** 100% solid phase  $\alpha$



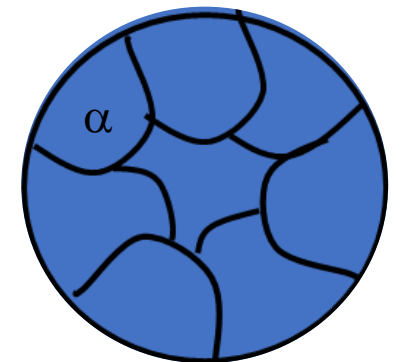


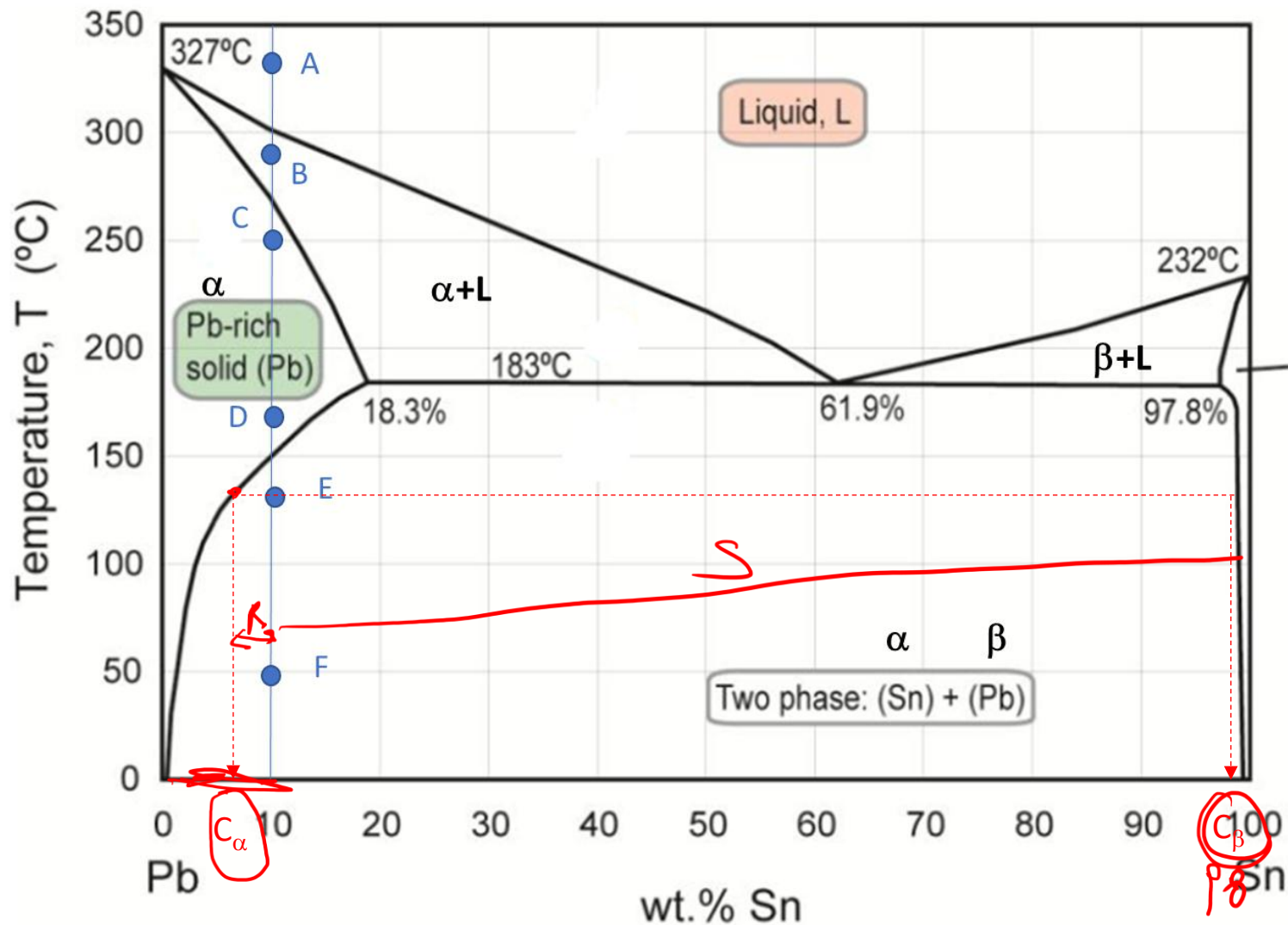
### Point D:

**Phases:** 1 phase solid ( $\alpha$ )

**Composition:** 10% Sn, 90% Pb

**Relative amount:** 100% solid phase  $\alpha$





### Point E:

**Phases:** 2 phases solid ( $\alpha$ ) + solid ( $\beta$ )

**Composition:** 2 phases region → horizontal rule

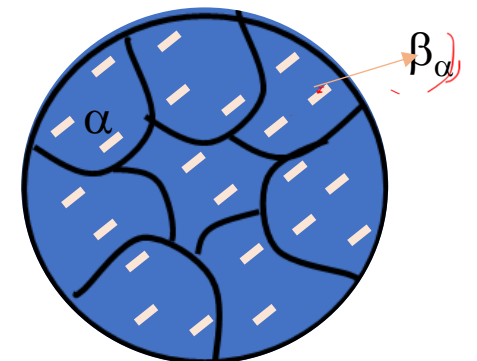
**Composition of solid ( $\alpha$ ):**  $C_\alpha =$  7%Sn, 83%Pb

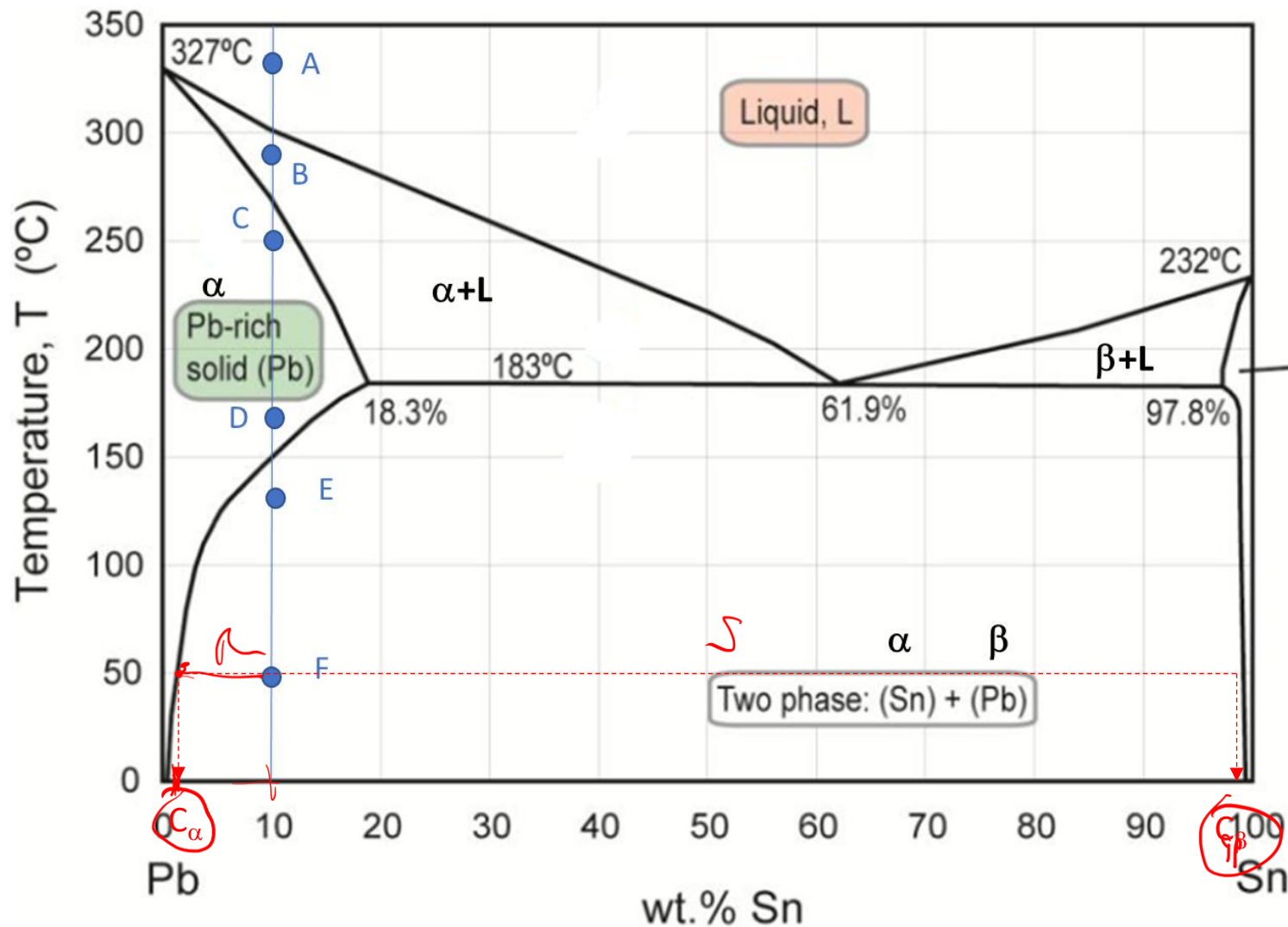
**Composition of solid ( $\beta$ ):**  $C_\beta =$  98%Sn, 2%Pb

**Relative amount:** 2 phases region → lever rule

$$\% \alpha = \frac{(98-10)}{(98-7)} \times 100 = 96.7\%$$

$$\% \beta = \frac{(10-7)}{(98-7)} \times 100 = 3.3\%$$





### Point F:

**Phases:** 2 phases solid ( $\alpha$ ) + solid ( $\beta$ )

**Composition:** 2 phases region  
→ horizontal rule

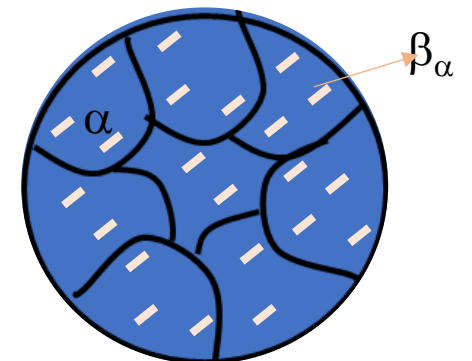
**Composition of solid ( $\alpha$ ):**  $C_{\alpha} =$   
2%Sn, 98%Pb

Composition of solid ( $\beta$ ):  $C_{\beta} =$   
99%Sn, 1%Pb

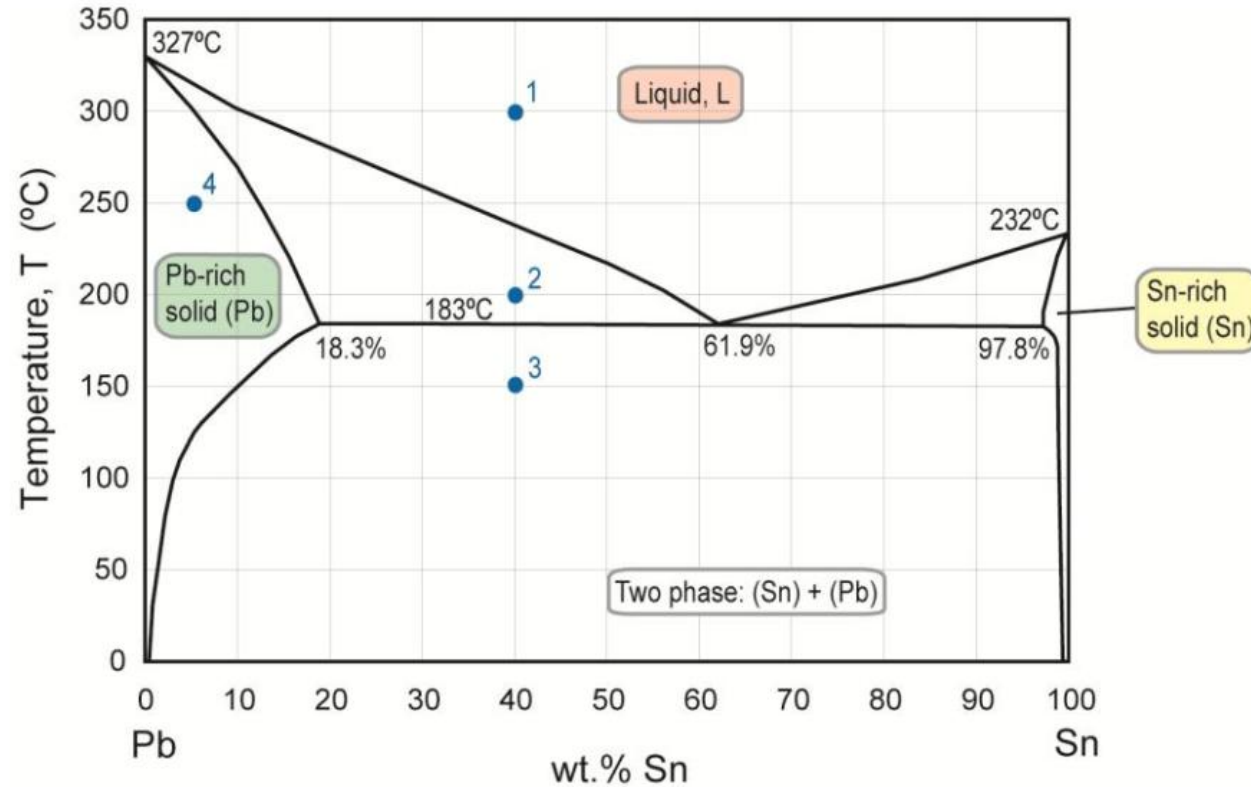
**Relative amount:** 2 phases  
region → lever rule

$$\% \alpha = \frac{(99-10)}{(99-2)} \times 100 = 91.7\%$$

$$\% \beta = \frac{(10-2)}{(99-2)} \times 100 = 8.3\%$$

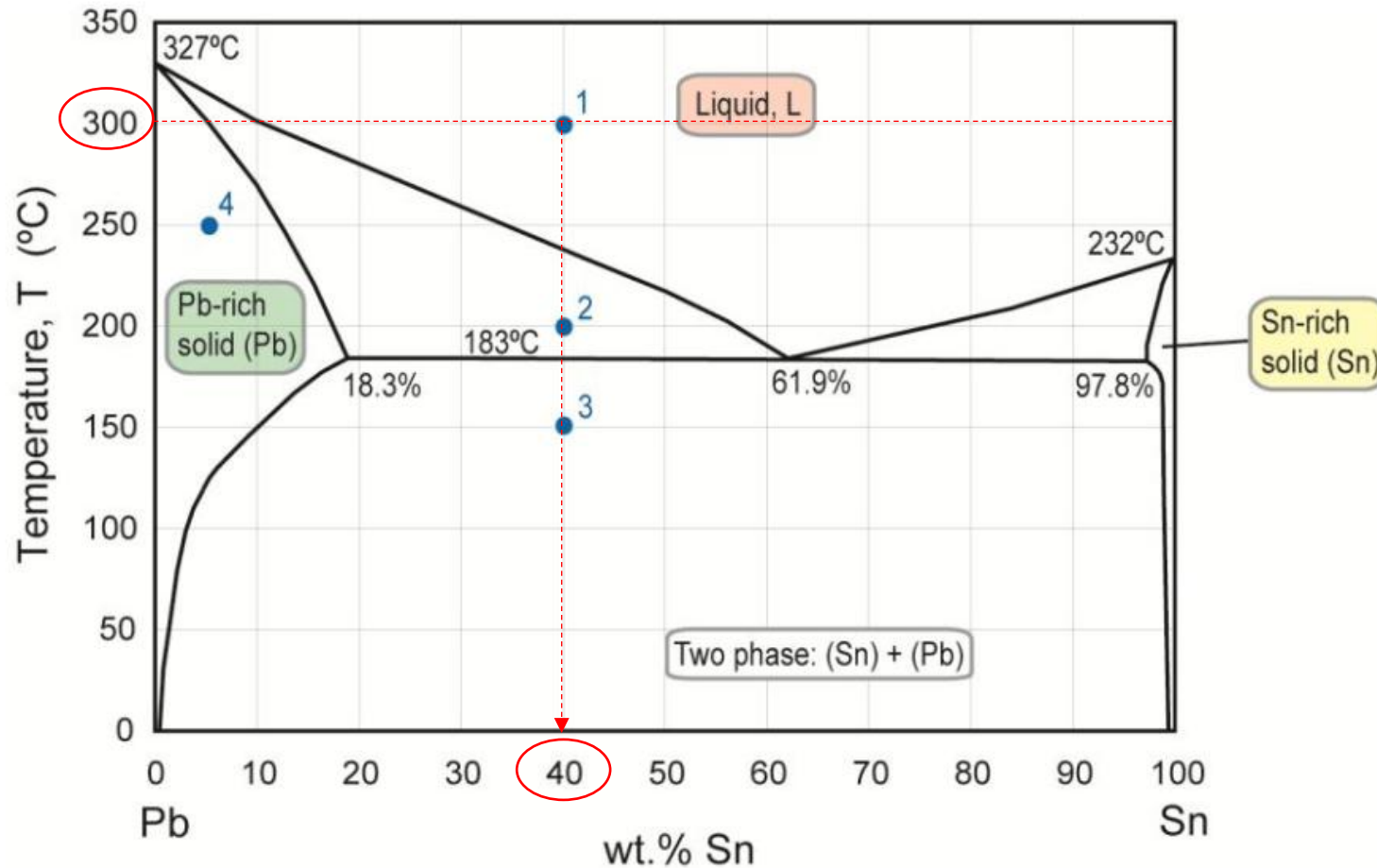


**E.6** Use the Pb-Sn diagram below to answer the following questions.



- (a) What are the values of the state variables (composition and temperature) at constitution point 1?
- (b) Mark the constitution points for Pb-60wt% Sn and Pb-20wt% Sn alloys at 250°C. What phases are present in each case?
- (c) The alloy at constitution point 1 is cooled very slowly to room temperature, maintaining equilibrium. At which temperatures do changes in the number or type of phases occur? What phases are present at constitution points 2 and 3?
- (d) The alloy at constitution point 4 is cooled slowly to room temperature. Identify the following:
  - (i) the initial composition, temperature and phase(s);
  - (ii) the temperature at which a phase change occurs, and the final phase(s).

(a) What are the values of the state variables (composition and temperature) at constitution point 1?



Point 1

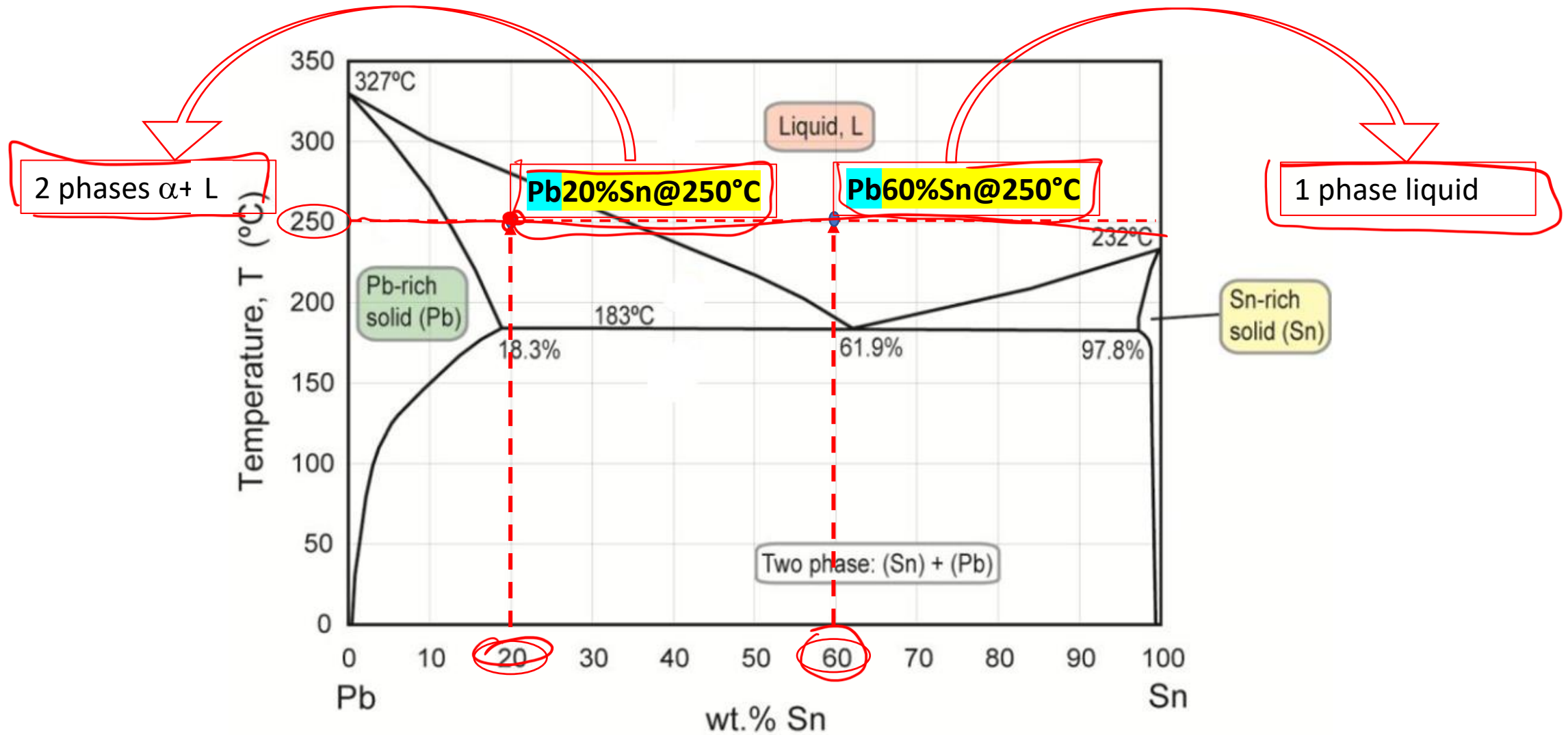
Phase:

1 phase liquid

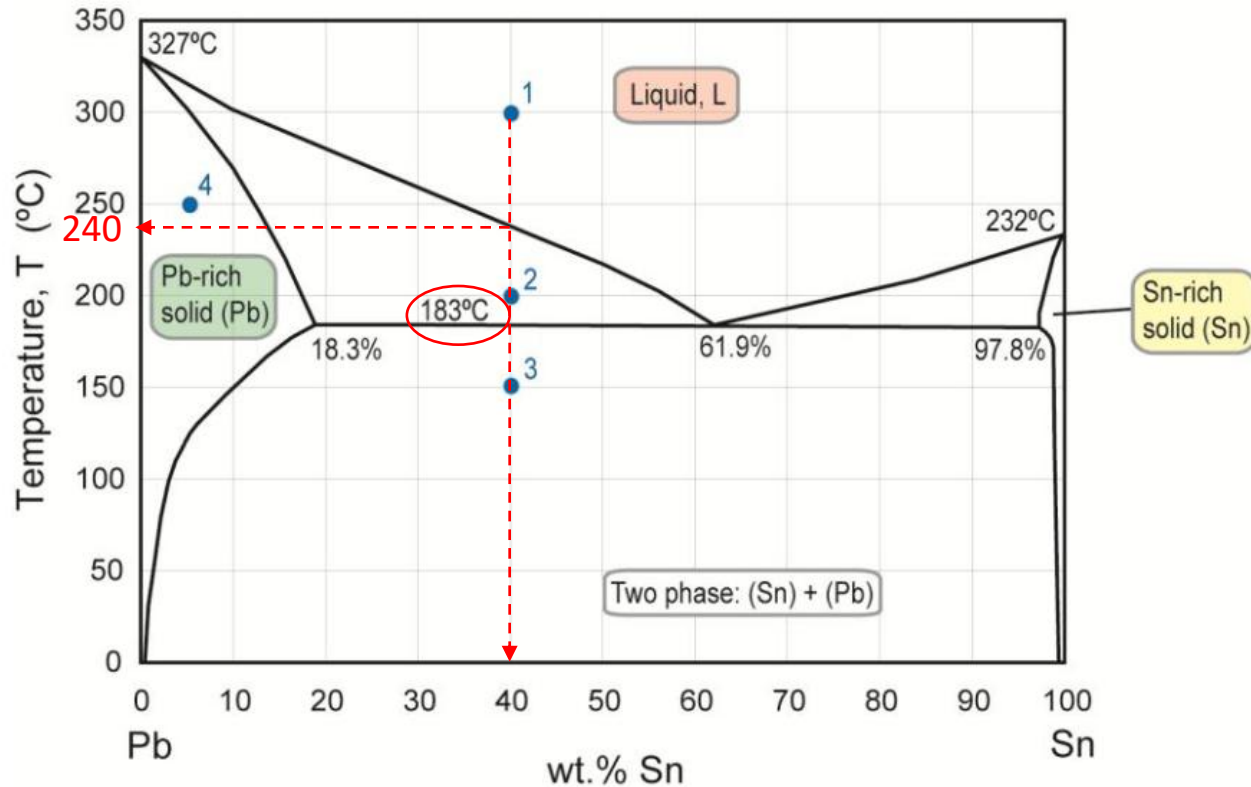
composition: 40%Sn, 60% Pb  $(100 - 40) = 60\%$   
Pb

Temperature: 300°C

(b) Mark the constitution points for Pb-60wt% Sn and Pb-20wt% Sn alloys at 250°C. What phases are present in each case?



(c) The alloy at constitution point 1 is cooled very slowly to room temperature, maintaining equilibrium. At which temperatures do changes in the number or type of phases occur? What phases are present at constitution points 2 and 3?



### Point 1:

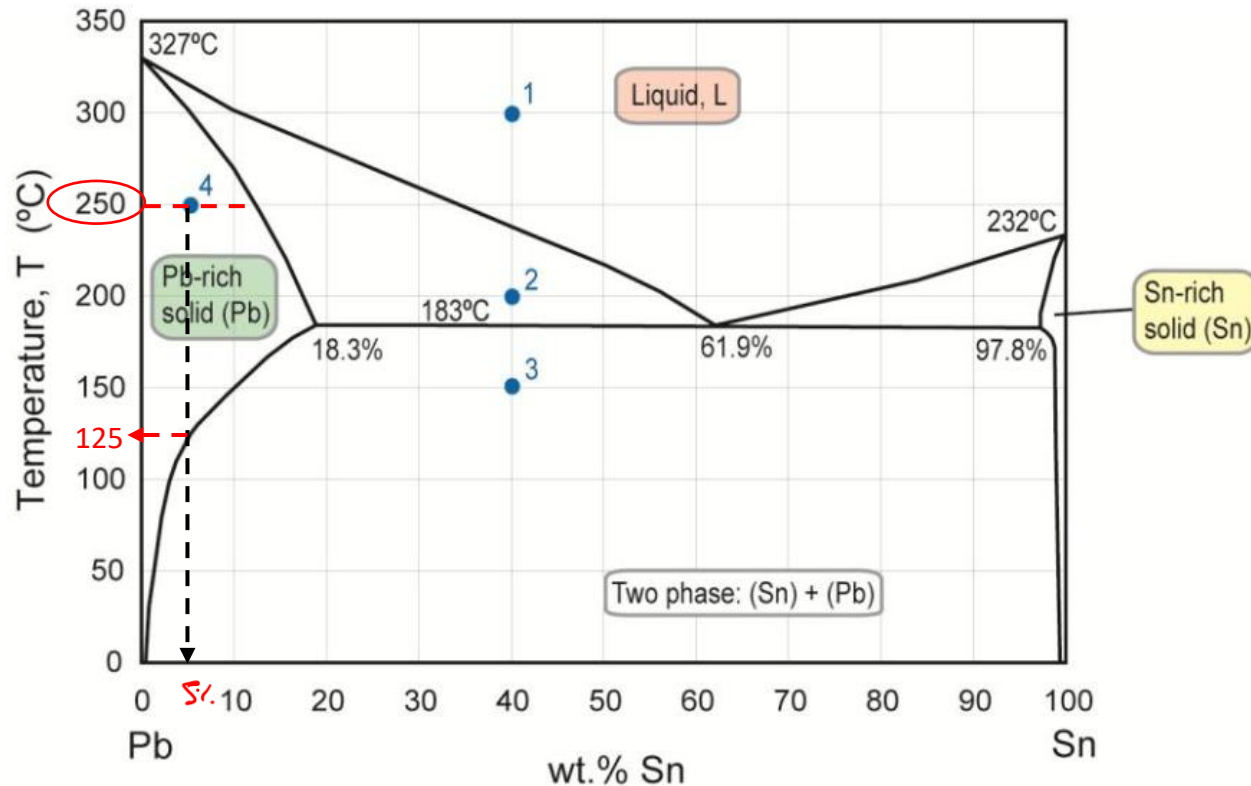
- At point 1 → we are at 300°C – phase: only liquid
- Proceed into the cooling → at 240°C - solidification of the first  $\alpha$  crystal
- At 183°C - all the liquid undergoes eutectic transformation producing  $\alpha$ - $\beta$  lamellar microstructure

**Point 2:** 2 phases  $\alpha$  (solid rich in Pb) and Liquid (L)

**Point 3:** 2 phases  $\alpha$  (solid rich in Pb) and  $\beta$  (solid rich in Sn)

(d) The alloy at constitution point 4 is cooled slowly to room temperature. Identify the following:

- (i) the initial composition, temperature and phase(s);
- (ii) the temperature at which a phase change occurs, and the final phase(s).

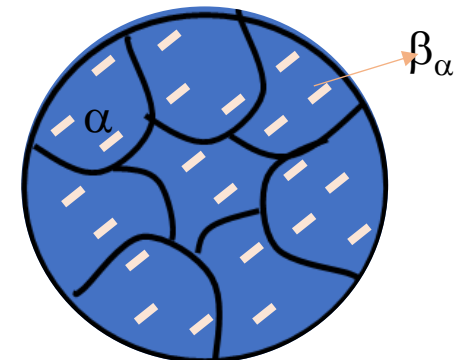


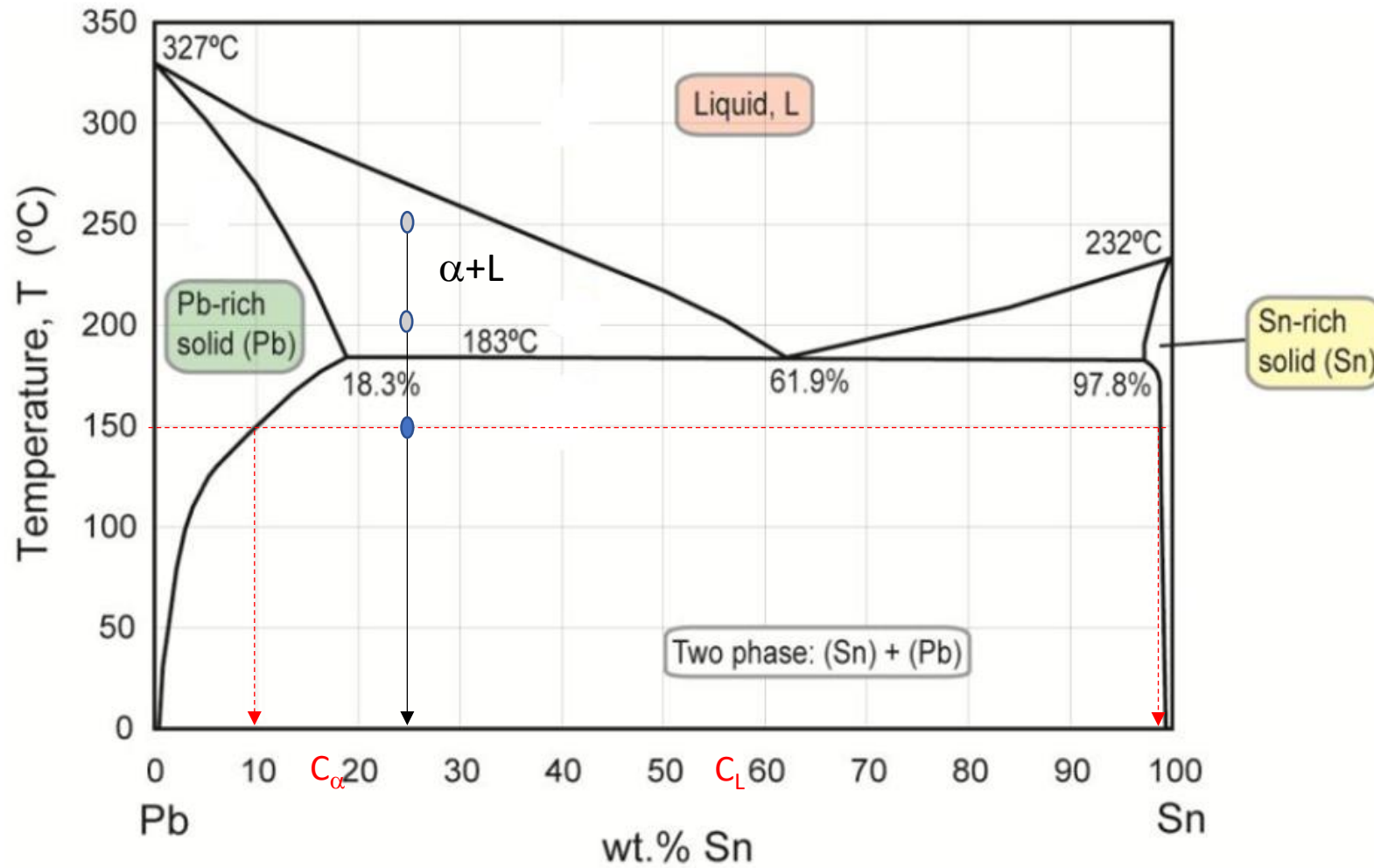
Point 4:

Start – 250°C,  $\alpha$  (solid phase rich in Pb),  
composition : 5%Sn, 95%Pb

125°C precipitation of  $\beta_{\alpha}$  solid phase (rich in Sn)  
due to reduction of the solubility of Sn in Pb at that  
temperature

From 125°C to room temperature the  
microstructure is solid phase  $\alpha$  (rich in Pb) with  
small  $\beta_{\alpha}$  crystals



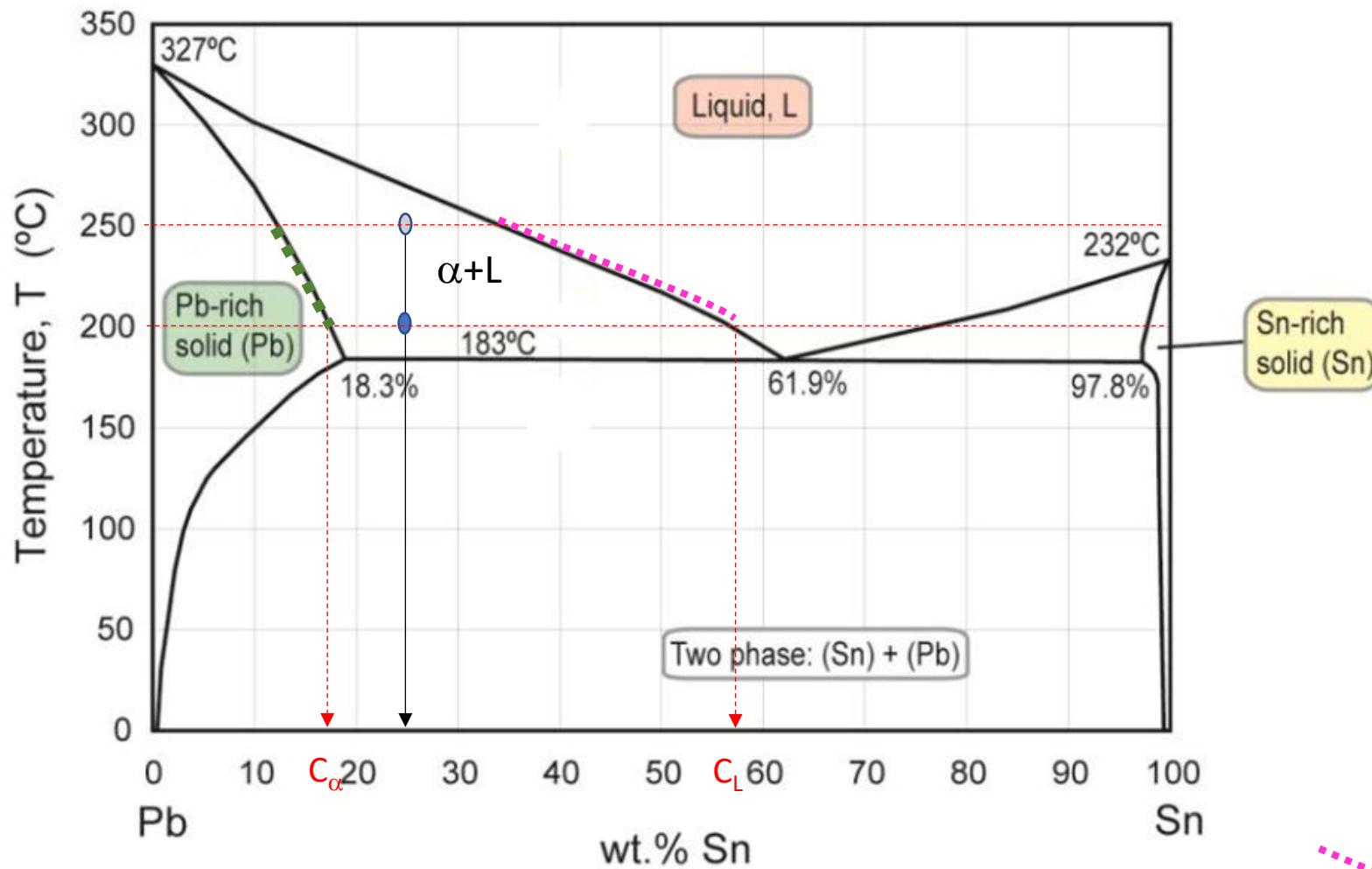


**Alloy 25%Sn 75%Pb @ 150 $^{\circ}\text{C}$**

$\alpha$  (solid phase rich in Pb) and  
 $\beta$  (solid rich in Sn)

Composition of  $\alpha$ : 10%Sn,  
 90%Pb

Composition of  $\beta$ : 99%Sn,  
 1%Pb



### Alloy 25%Sn 75%Pb @ 250°C

α (solid phase rich in Pb) and liquid (L)

Composition of α = 12%Sn, 88%Pb

Composition of liquid: 33%Sn, 67%Pb

### Alloy 25%Sn 75%Pb @ 200°C

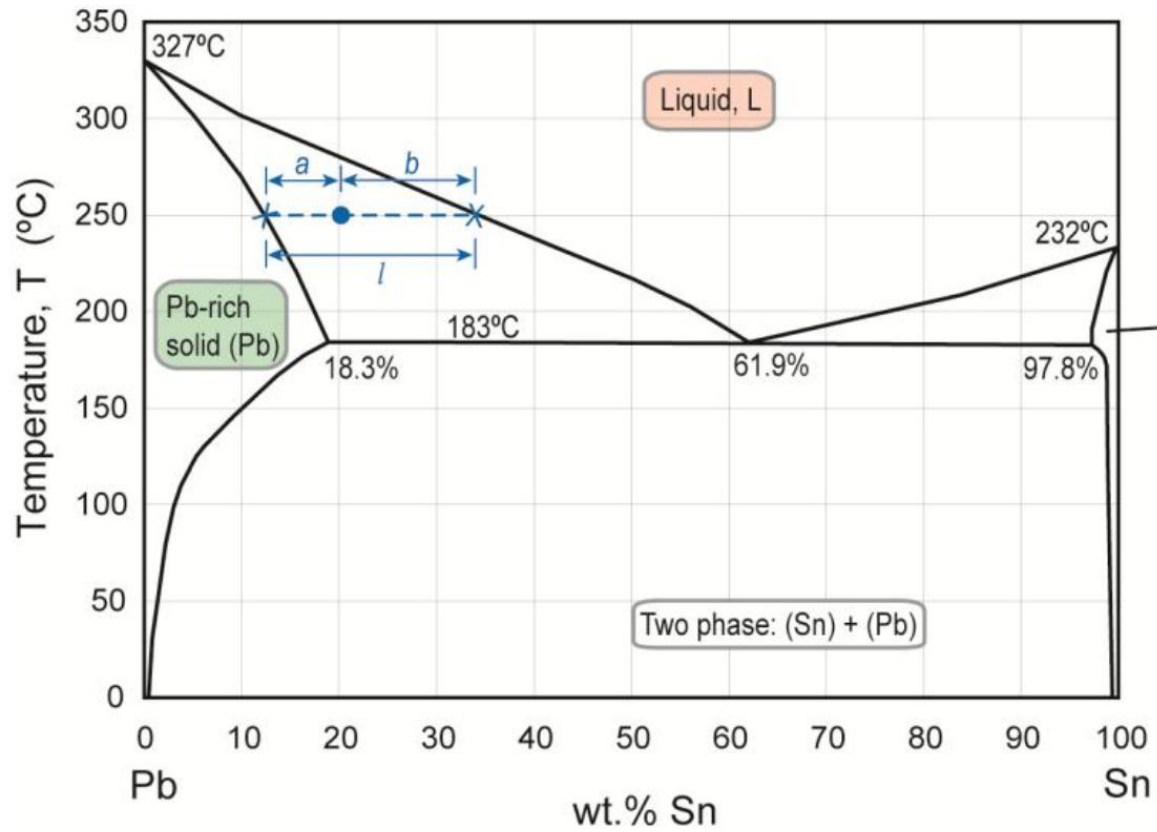
α (solid phase rich in Pb) and liquid (L)

Composition of α = 18%Sn, 82%Pb

Composition of liquid: 58%Sn, 42%Pb

Variation of α composition reducing temperature from 250 to 200 °C

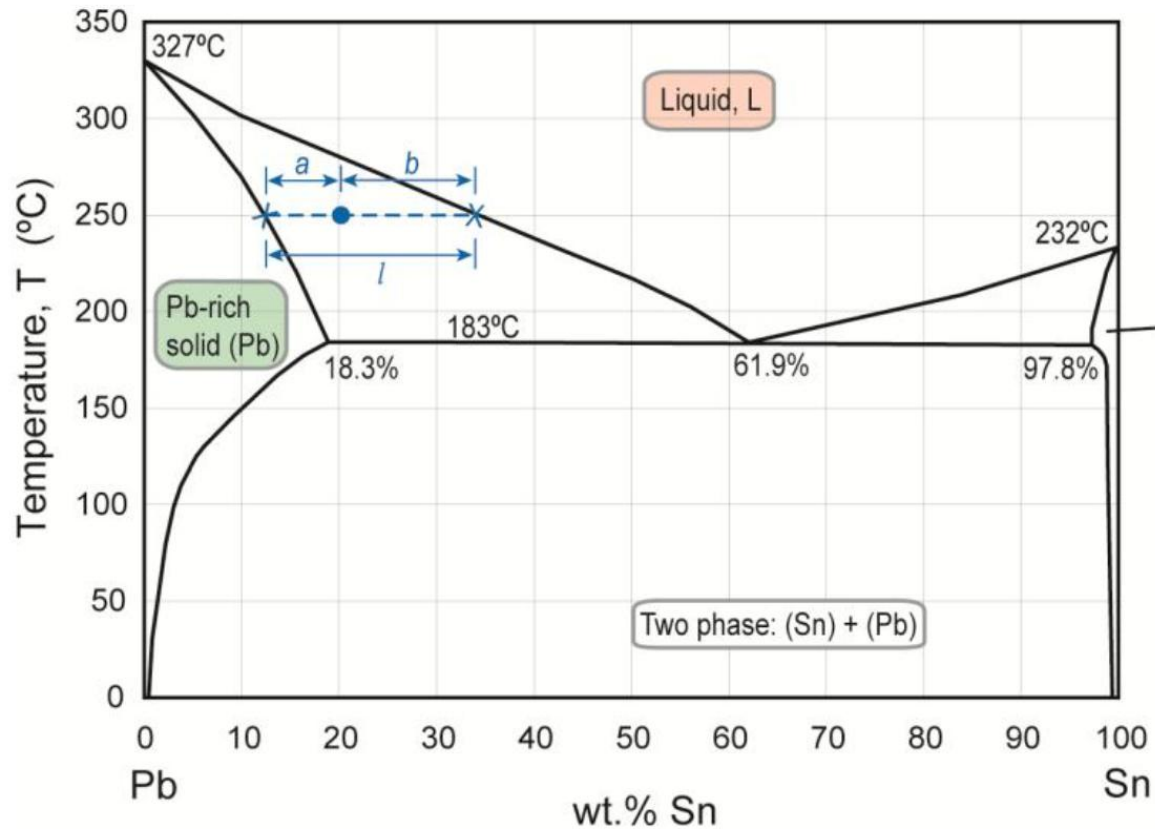
Variation of L composition reducing temperature from 250 to 200 °C



Consider Pb-20wt% Sn at 250  $^{\circ}\text{C}$  in Figure

A constitution point in the two-phase field: liquid plus Pb-rich solid.

**Find the proportions of each phase**



Consider Pb-20wt% Sn at 250 °C in Figure

A constitution point in the two-phase field: liquid plus Pb-rich solid.

**To find the proportions of each phase**

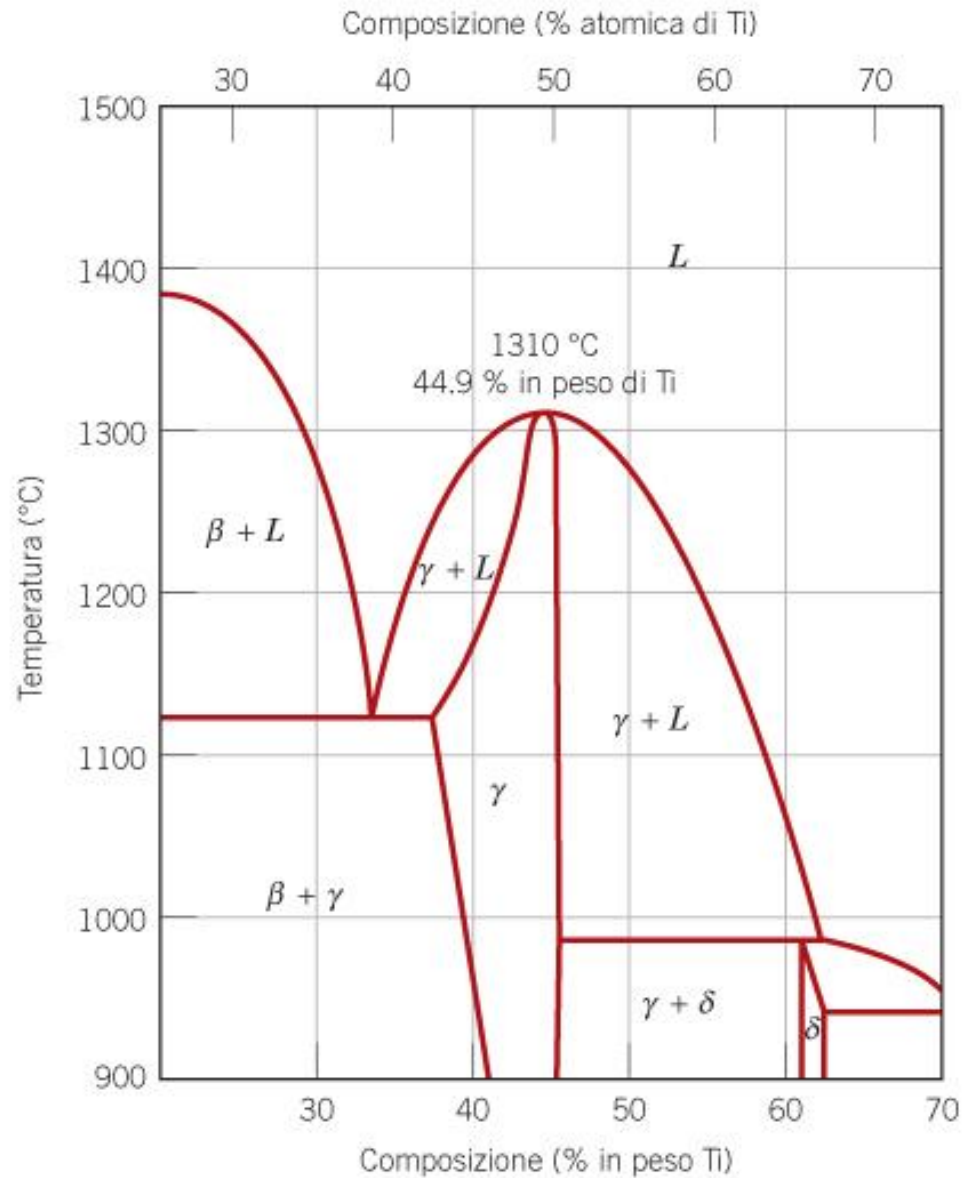
First construct a tie-line through the constitution point and read off the compositions of the phases:

*Pb-rich solid* with  $W_{Sn}^{SOL} = 12\%$ ; *Liquid* with  $W_{Sn}^{LIQ} = 34\%$ .

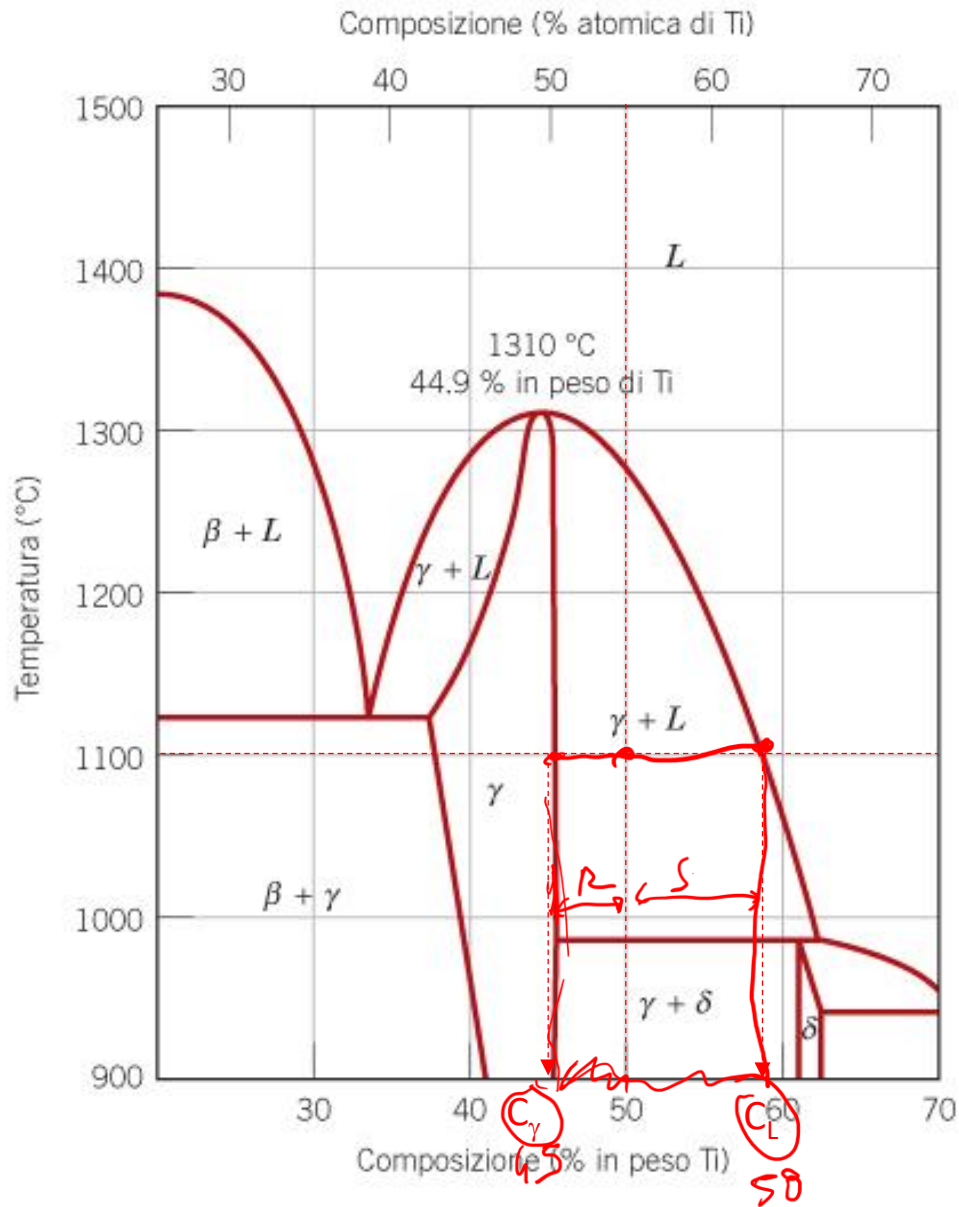
The tie-line is of length  $l$ , while the lengths of the segments to either side of the constitution point are **a** and **b** respectively

For the example alloy of composition  $W_{Sn} = 20\%$

Determine the phase/s present, the composition/s and relative amount/s for an alloy with 50%Ti at 1100°C.



Determine the phase/s present, the composition/s and relative amount/s for an alloy with 50%Ti at 1100°C.



Phases: solid phase  $\gamma$  (intermediate solid solution) and liquid L

Compositions:

Composition of  $\gamma$ : 45%Ti

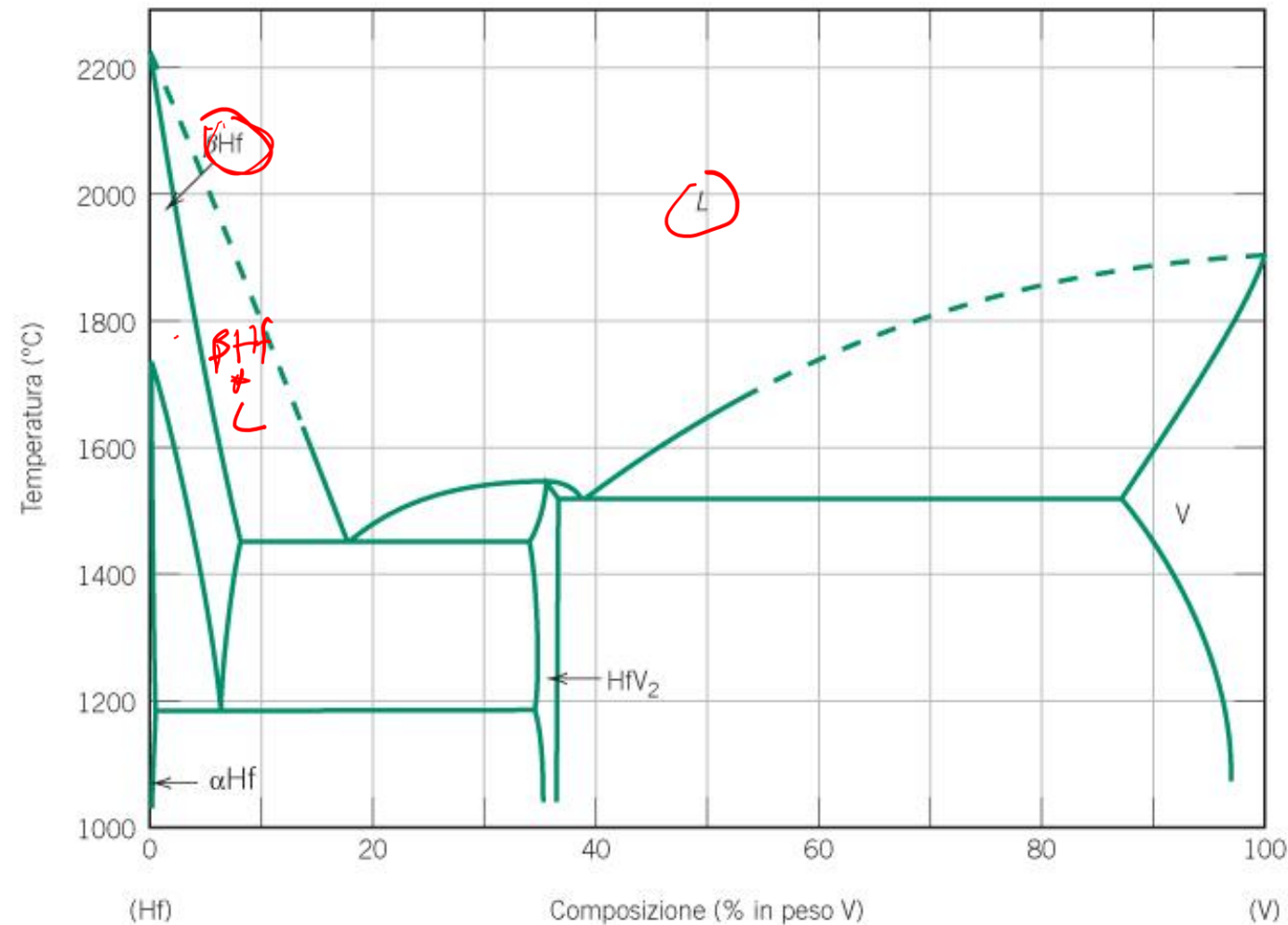
Composition of Liquid: 58%Ti

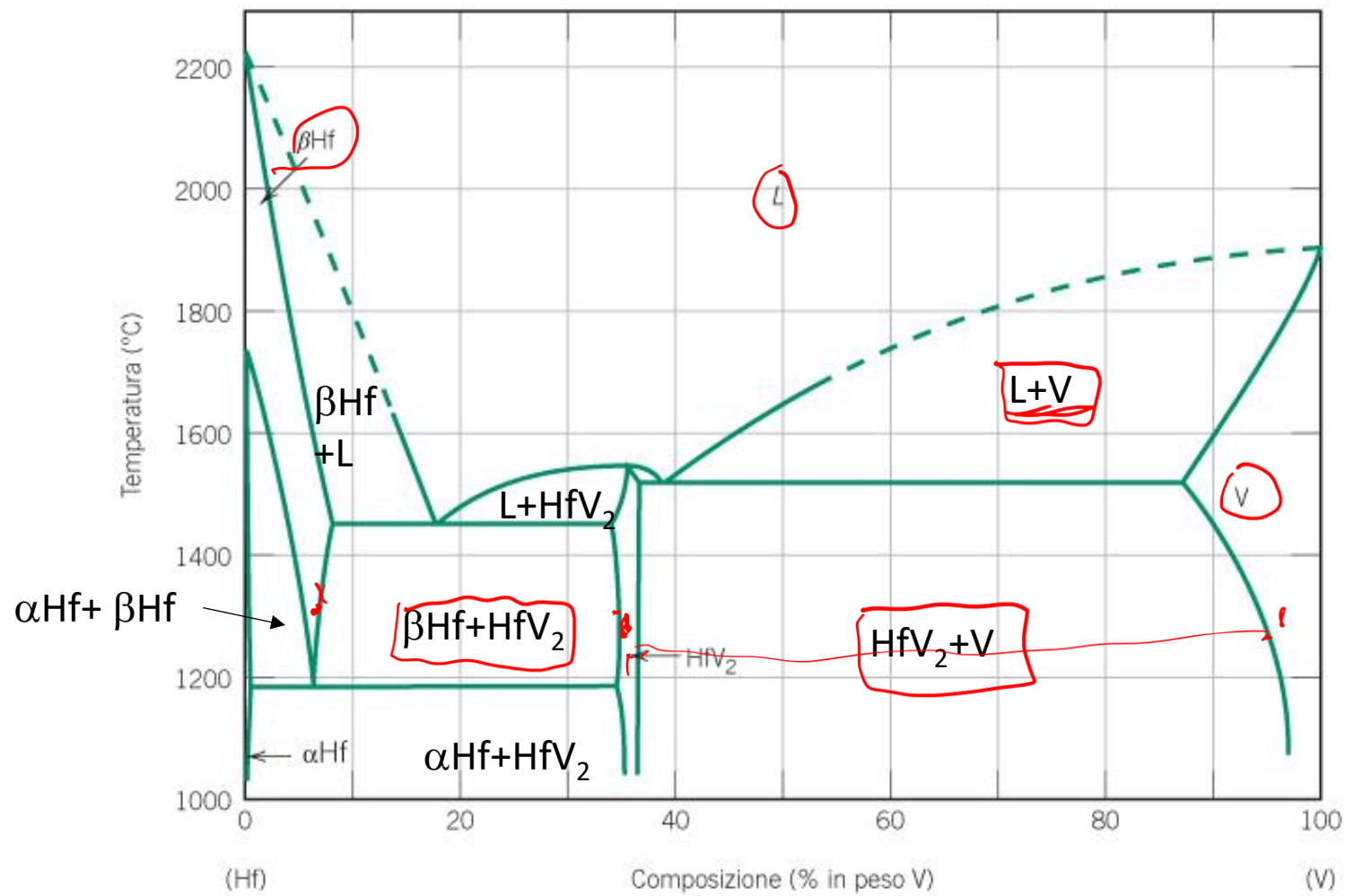
Relative amounts

$$\% \gamma = \frac{(58-50)}{(58-45)} \times 100 = 61.5\%$$

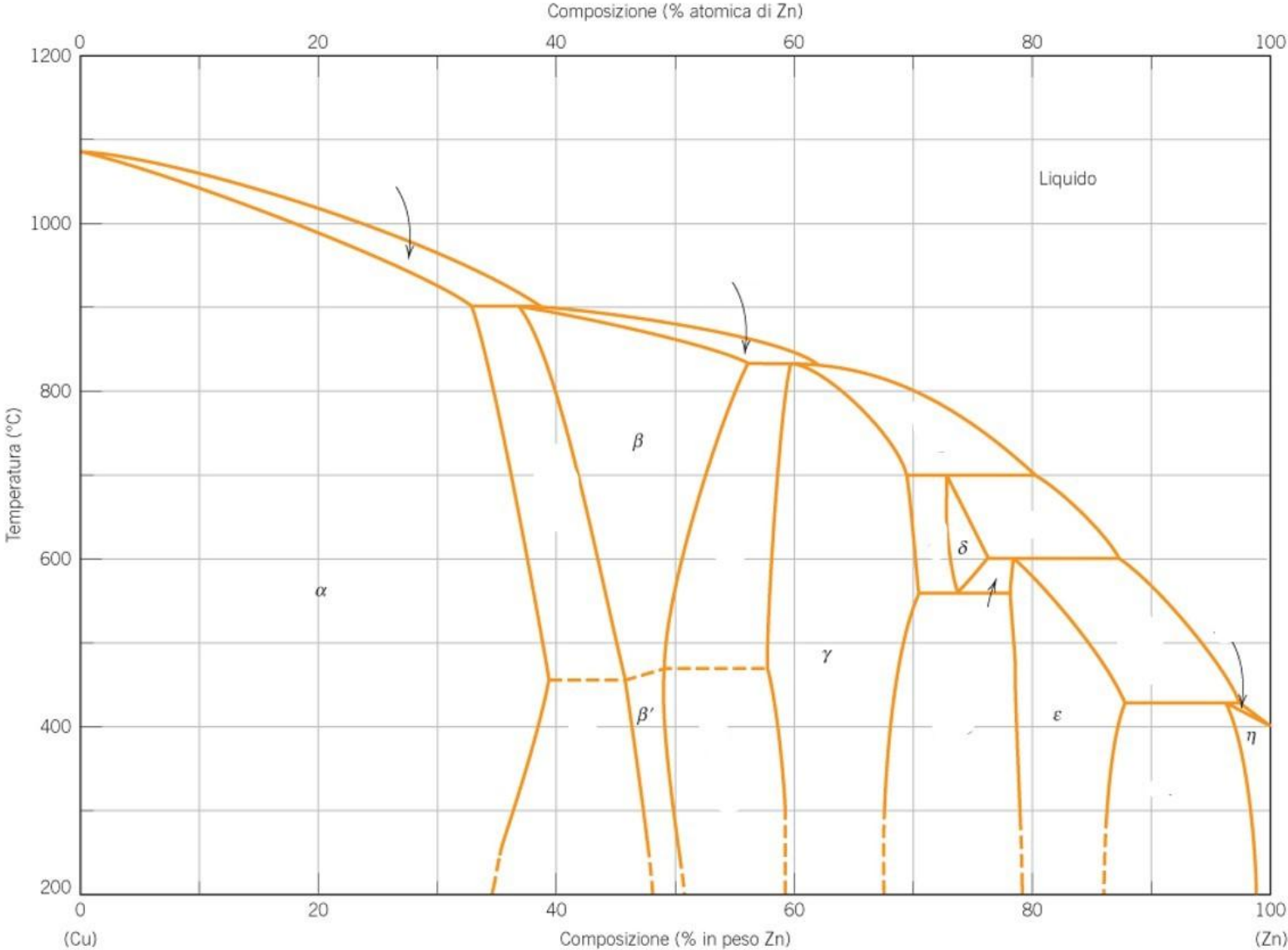
$$\% L = \frac{(50-45)}{(58-45)} \times 100 = 38.5\%$$

Complete the following diagram





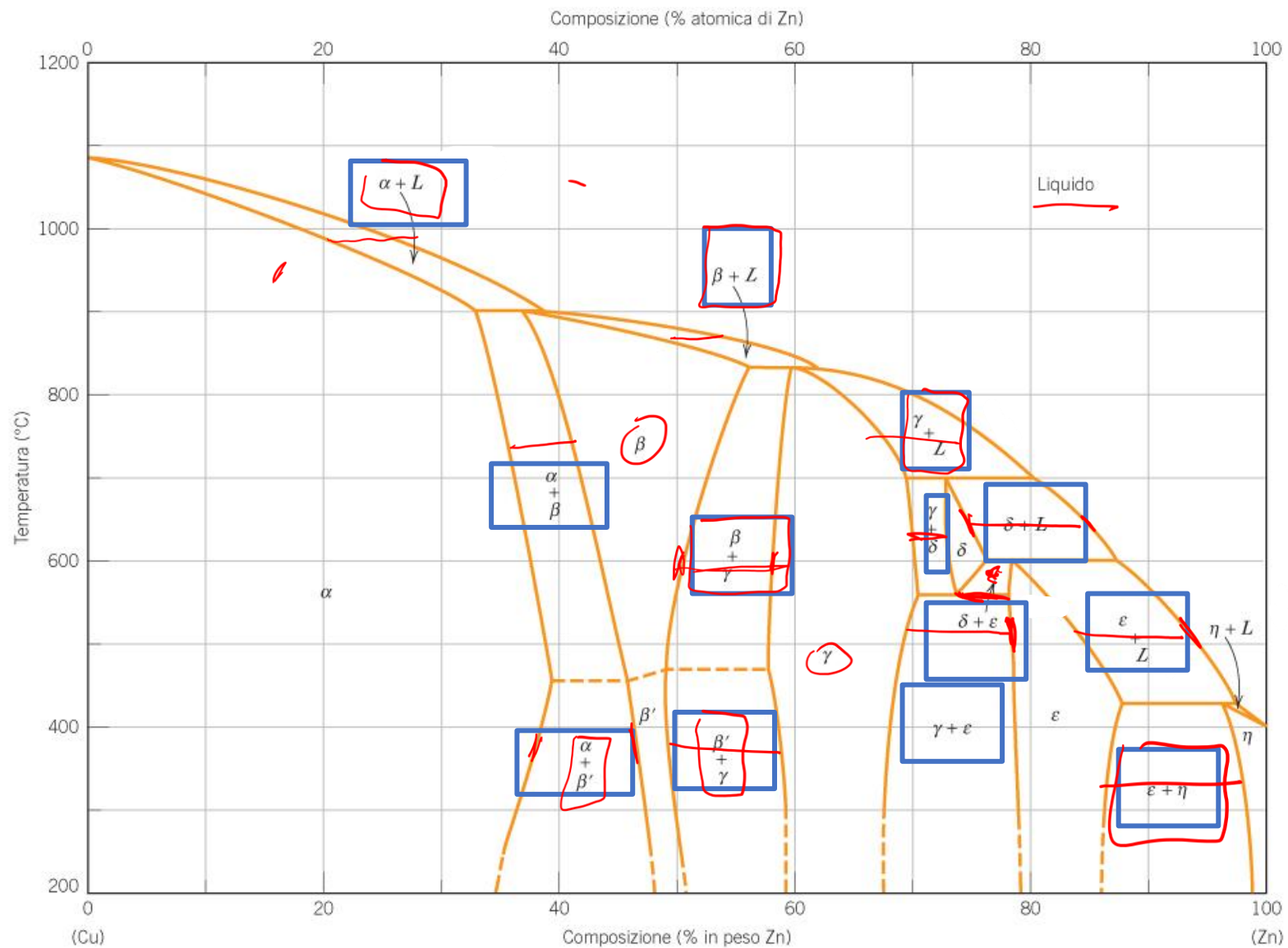
Complete the following diagram



**Figura 9.19** Diagramma di fase rame-zinco.



William D. Callister Jr., David G. Rethwisch  
Scienza ed Ingegneria dei Materiali  
EdiSES



**Figura 9.19** Diagramma di fase rame-zinco.